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Testing. Advising. Assuring.

Title:

The Fire Resistance Performance of Two-Panel, Side and Centre-Opening 'Compact' Lift Landing Doorsets

WF Assessment Report No:

149621 (Issue 2)

Prepared for:

Tecnolama SA

Ctra. De Constanti
Km. 3
43206 Reus
Spain

Date: 15th December 2005

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Executive Summary

Objective	This report provides an appraisal of the fire resistance performance of two-panel, side and centre-opening 'Compact' lift landing doorsets, which are similar to previously fire tested assemblies.
Report Sponsor	Tecnolama SA
Address	Ctra. De Constanti Km. 3 43206 Reus Spain
Summary of Conclusions	It can be concluded that the proposed doorsets would be expected to provide 120 minutes integrity performance (and therefore the required E120 classification) if subjected to fire resistance tests in accordance with BS EN 81-58: 2003.
Valid until	1st January 2016

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Introduction

This report provides an appraisal of the fire resistance performance of two-panel side and centre-opening 'Compact' lift landing doorsets.

FTSG

The data referred to in the supporting data section has been considered for the purpose of this appraisal which has been prepared in accordance with the Fire Test Study Group Resolution No. 82: 2001.

Assumptions

It is assumed that the wall constructions to which the doorsets will be mounted will be of reinforced concrete or masonry, capable of providing effective structural support for the required period.

It is also assumed that the general construction of the doorsets and the materials used in the construction will, unless specifically detailed in this report, be identical to those of the tested assemblies.

It is further assumed that the doorsets will be required to provide fire resistance when the landing side only is exposed to the heating conditions of a fire resistance test.

It is assumed that the doorsets will be installed by competent installers in a similar manner to that used when installing the fire tested assembly.

Proposals

The construction of the proposed doorsets shall be closely based upon the assembly previously fire tested under the reference WF Report No. 146516. It is, however, proposed that the doorset may incorporate various changes which are summarised as follows:

a) The doorset may be used in a two-panel, side-opening configuration in addition to the tested centre-opening configuration.

b) The doorset may incorporate various side-panel, control box, LDU and frame post configurations, shown on the drawings appended to this report.

c) The clear opening size of both the side and centre-opening doorsets may be between 2000 and 2300 mm high and 600 and 1100 mm wide.

d) The door frame may be installed in any of the following three ways:

- Inside the reveal of the opening with an adjustable steel profile surrounding the perimeter
- Inside the reveal of the opening with the perimeter of the doorset infilled with concrete.
- Face fixed to the 'shaft' face of the wall

Basic Test Evidence

WF Report No. 146516

A fire resistance test conducted in accordance with BS EN 81-58: 2003, on a specimen of an uninsulated, two-panel, centre-opening, lift landing doorset.

The doorset had clear opening dimensions of 2000 mm high by 900 mm wide and incorporated a side-panel and LDU adjacent to one side of the doorset and a differently constructed side-panel and control box adjacent to the other.

The specimen satisfied the integrity performance requirements of the Standard for a period of 120 minutes after which time the test was discontinued.

Assessed Performance

Side-Opening Doorsets

The original fire test referenced 146516 was performed on a specimen of a two-panel, centre-opening lift landing doorset.

A separate fire test has been conducted on a similarly constructed side-opening doorset under the reference WF Report No. 146523. This tested doorset incorporated insulation panels on the unexposed face of the door panels and frame, but the general construction of the doorset was otherwise the same as the tested centre-opening doorset.

This test was conducted on a doorset having a clear opening size of 2000 mm high by 900 mm wide, as proposed, and incorporated a side-panel on each side of the doorset. The test was conducted in accordance with BS EN 81-58: 2003 and the specimen provided 120 minutes integrity performance.

In many respects the incorporation of these insulation panels on the unexposed face of the doorset is expected to have had a detrimental influence on the integrity performance, since these will have prevented dissipation of the heat from the unexposed face of the doorset. In turn this will have left the steel panels subjected to higher temperatures, further weakening the doorset and leaving it prone to increased distortion.

Confidence can, however, be gained from this test since the maximum recorded leakage throughout the required 120 minute period was only 1.62 m^3/(min.m), just over 50% of that required to cause integrity failure.

Based upon the proven fire performance of the uninsulated, centre-opening doorset, along with additional test evidence to support the fire performance of a similarly constructed, but insulated, side-opening doorset, it is expected that the proposed uninsulated side-opening doorset should be capable of providing the required 120 minute fire performance.

Proposed Configurations

The proposal requires the use of different side-panel, control box, LDU and frame post configurations adjacent to the doorsets, as shown in the drawings appended to this report.

Each of the proposed items was previously included on the doorset tested under the reference WF No. 146516 without detriment to the fire performance. The test specimen was designed to represent a relatively onerous configuration in order to maximise the coverage for assessment purposes. The proposed doorsets may not necessarily incorporate all of the proposed options simultaneously, but various combinations of these tested items are instead proposed.

The only notable deviation from the tested assembly is where the side frame may be adapted for situations in which it is not required to be used as an LDU. In such a case, the changes to the tested frame post are minimal and are not expected to have any discernible influence on the fire performance.

The tested doorset incorporated side-panels of 650 mm width, which it is proposed may be increased to a maximum of 985 mm. Due to the restrictions on furnace size, it is not practical to test large side-panel dimensions whilst still retaining a reasonable clear opening size. The tested 650 mm wide panel is, however, deemed to have provided a reasonable representation of the proposed range of dimensions and a further increase in width by 335 mm as proposed is not expected to detract from the fire performance.

For the side-opening doorsets, the configurations shall comprise a combination of the options successfully tested under the references WF Nos. 146523 and 146516, with the slam post being adapted slightly for situations where this is not required to be used as an LDU. The changes to the profile of this frame section are minor and are not expected to have an influence on the fire performance.

All proposed configurations may therefore be positively appraised.

Clear Opening Size

The tested centre-opening doorset had a clear opening size of 2000 mm high by 900 mm wide.

According to the Direct Field of Application provided within BS EN 81-58: 2003, the doorset may be increased in height by up to 15% and the width of the doorset may be increased or reduced by up to 30%. The proposed clear opening height ranges between 2000 mm and 2300 mm therefore falls inside with the requirements of the Direct Field of Application. The maximum proposed width of 1100 mm (a 22% increase in width) also falls within the permissible size variation, although the maximum reduction in width of 33% to 600 mm, marginally exceeds that permissible by the Standard.

It is, however, expected that, considering the exceptional performance of the tested side and centre-opening doorsets, the maximum leakage recorded in both tests being only marginally in excess of 2 m^3/(min.m), the additional reduction in size by up to 3% should not detract from the required performance.

The proposed range of clear opening sizes may therefore be positively appraised.

Installation

The doorset successfully tested under the reference WF No. 146516 was installed inside the clear opening with an adjustable steel profile incorporated between the frame sections and the wall to prevent leakage during the test and the formation of through gaps.

This proposed method of installation has therefore proven to be suitable for use.

A variation on this method of installation is where the doorsets are installed in a similar manner but without the adjustable steel profile and the interface between the frame and the wall is then infilled with concrete or mortar. The nominal gap between the frame and the wall shall be 10 mm and the concrete/mortar infill shall be applied through the full thickness of the frame.

The sealing of the perimeter of the doorset in this manner should provide an improved performance with respect to leakage. Although the tested adjustable profile shall abut the wall construction there may still be some small through gaps between the profile and the wall, depending upon the flatness of the wall.

The use of a concrete or mortar backfill should, however, ensure a tight seal between the wall and the doorset which can only enhance the integrity performance of the doorset.

The final option is for the doorset to be mounted to the ‘shaft’ face of the wall construction. This method of installation shall ensure that the frame sections are partially fire protected by the wall construction which can only improve their stability under fire test conditions.

This proposed method of installation is therefore also considered to be slightly advantageous compared with that tested and may be positively appraised.

Conclusions

It is expected that the side and centre-opening 'Compact' doorsets discussed within this report, should be capable of providing 120 minutes integrity performance if subjected to a fire resistance test in accordance with BS EN 81-58: 2003.

Validity

This assessment is issued on the basis of test data and information available at the time of issue. If contradictory evidence becomes available to Exova Warringtonfire the assessment will be unconditionally withdrawn and Tecnolama SA will be notified in writing. Similarly the assessment is invalidated if the assessed construction is subsequently tested because actual test data is deemed to take precedence over an expressed opinion. The assessment is valid initially for a period of five years i.e. until 1st January 2016, after which time it is recommended that it be returned for re-appraisal.

The appraisal is only valid provided that no other modifications are made to the tested construction other than those described in this report.

Summary of Primary Supporting Data

WF Report No. 146516

A report on a fire resistance test performed in accordance with BS EN 81-58: 2003, on a two panel, centre opening, lift landing doorset.

The doorset, which incorporated two mild steel door panels, was installed over an aperture within a brickwork wall of overall nominal dimensions 2190 mm high by 2200 mm wide to give a clear opening size of 2000 mm high by 900 mm wide. The specimen incorporated a fixed infill panel adjacent to each of the door panels and a control box was incorporated on one side of the doorset only, behind the infill panel. The doorset was faced fixed to the unexposed surface of the wall such that the landing side was exposed to the heating conditions of the test.

The specimen satisfied the integrity performance requirements for a period of 120 minutes

Test Date : 27th June 2005

Test Sponsor : Tecnolama SA

WF Report No. 146523

A report on a fire resistance test performed in accordance with BS EN 81-58: 2003, on a two panel, side opening, lift landing doorset.

The doorset, which incorporated two mild steel door panels, was installed over an aperture within a brickwork wall of overall nominal dimensions 2190 mm high by 2200 mm wide to give a clear opening size of 2000 mm high by 900 mm wide. The specimen incorporated a fixed infill panel adjacent to each of the door panels and a control box was incorporated on one side of the doorset only, behind the infill panel. The doorset was faced fixed to the unexposed surface of the wall such that the landing side was exposed to the heating conditions of the test.

The specimen satisfied the integrity performance requirements for a period of 120 minutes and the insulation performance requirements for a period of 61 minutes.

Test Date : 1st July 2005

Test Sponsor : Tecnolama SA

Declaration by Tecnolama SA

We the undersigned confirm that we have read and complied with the obligations placed on us by the UK Fire Test Study Group Resolution No. 82: 2001.

We confirm that the component or element of structure, which is the subject of this assessment, has not to our knowledge been subjected to a fire test to the Standard against which the assessment is being made.

We agree to withdraw this assessment from circulation should the component or element of structure be the subject of a fire test to the Standard against which this assessment is being made.

We are not aware of any information that could adversely affect the conclusions of this assessment.

If we subsequently become aware of any such information we agree to cease using the assessment and ask Exova Warringtonfire to withdraw the assessment.

Signed:

For and on behalf of:

Signatories

Responsible Officer

D Hankinson* - Technical Consultant

Approved

C Johnson* - Technical Officer

* For and on behalf of Exova Warringtonfire

Report Issued: 15th December 2005

Issue 2: Change to infill panel dimensions and revalidation (16th December 2010)

The assessment report is not valid unless it incorporates the declaration duly signed by the applicant.

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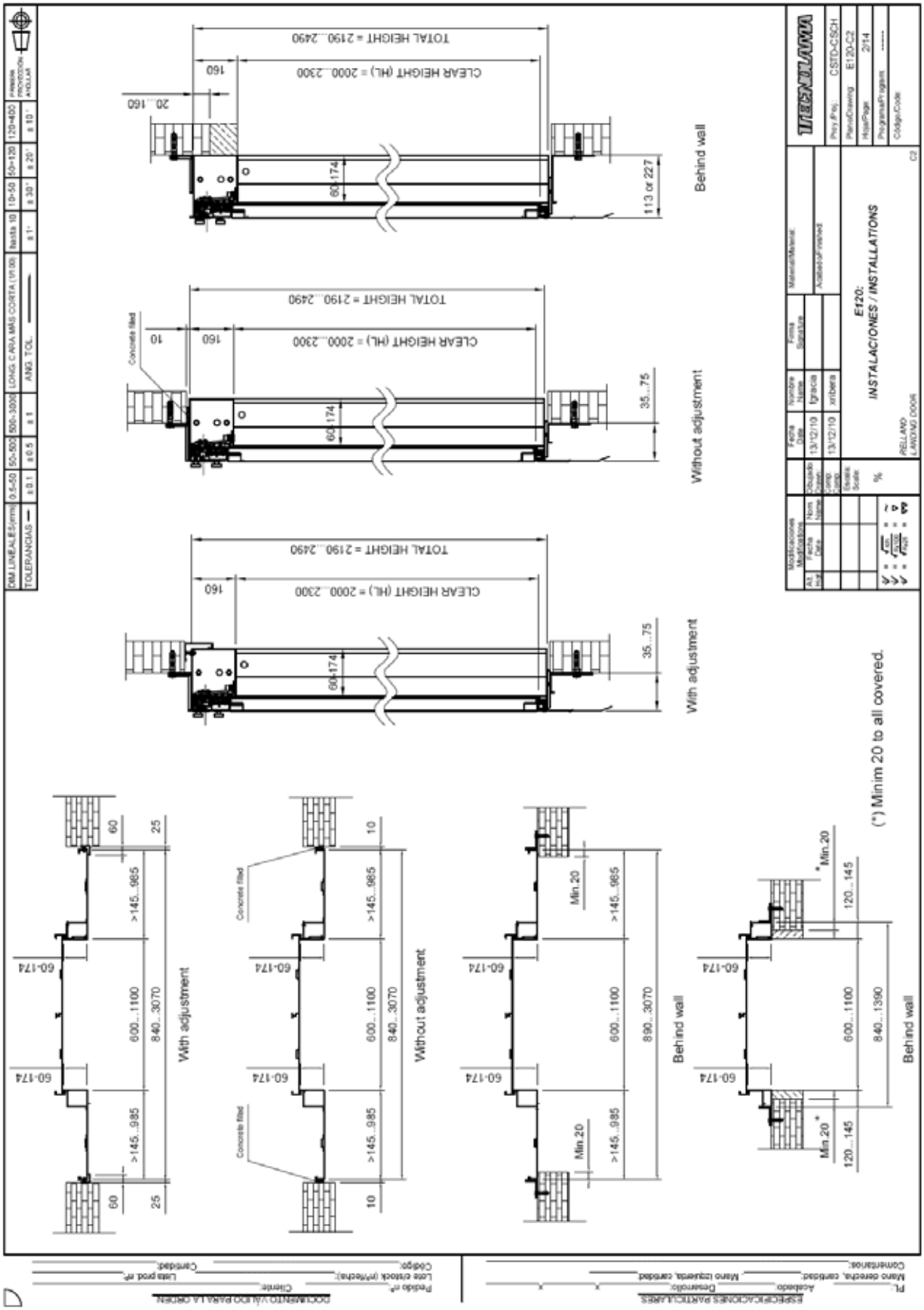
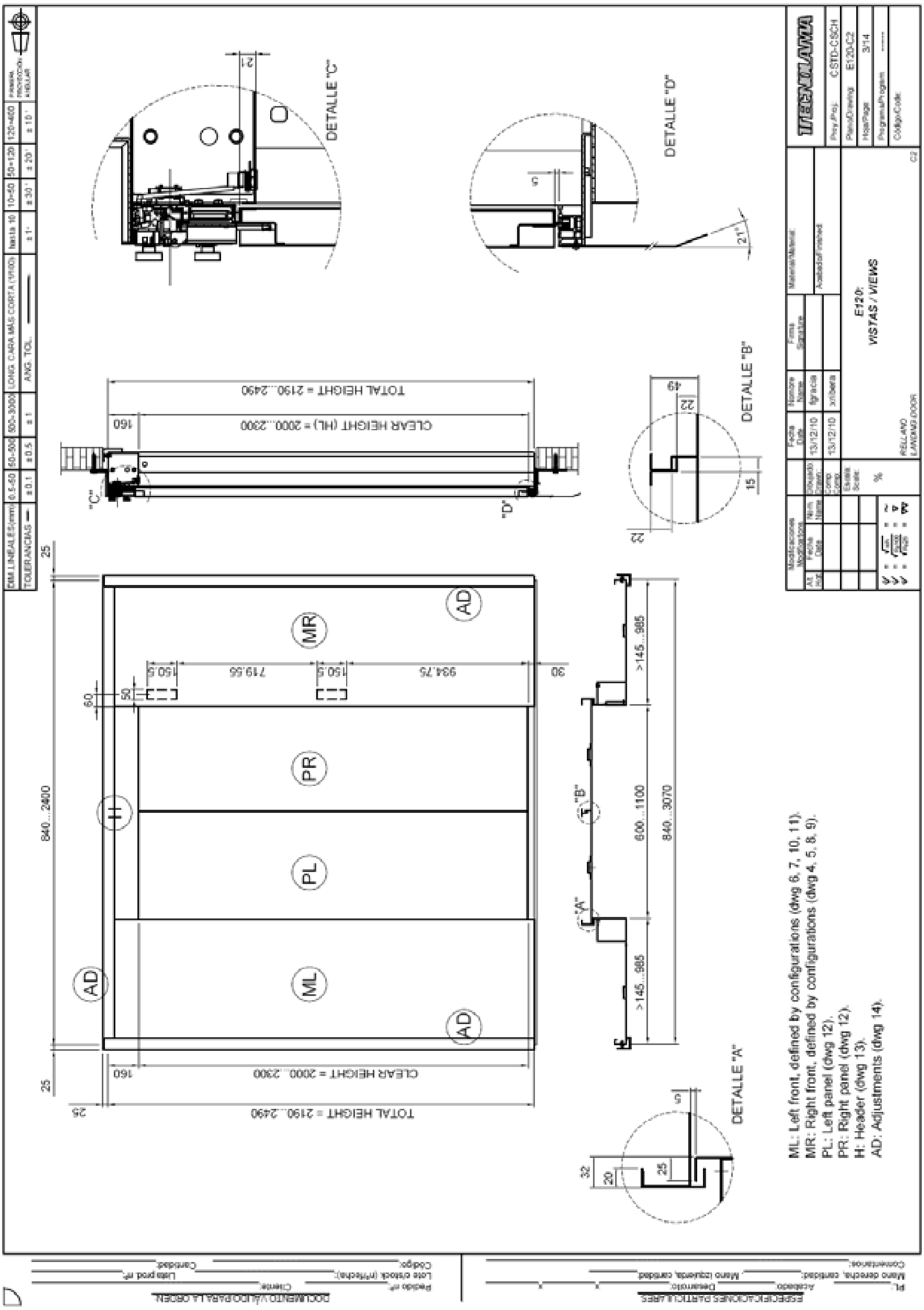
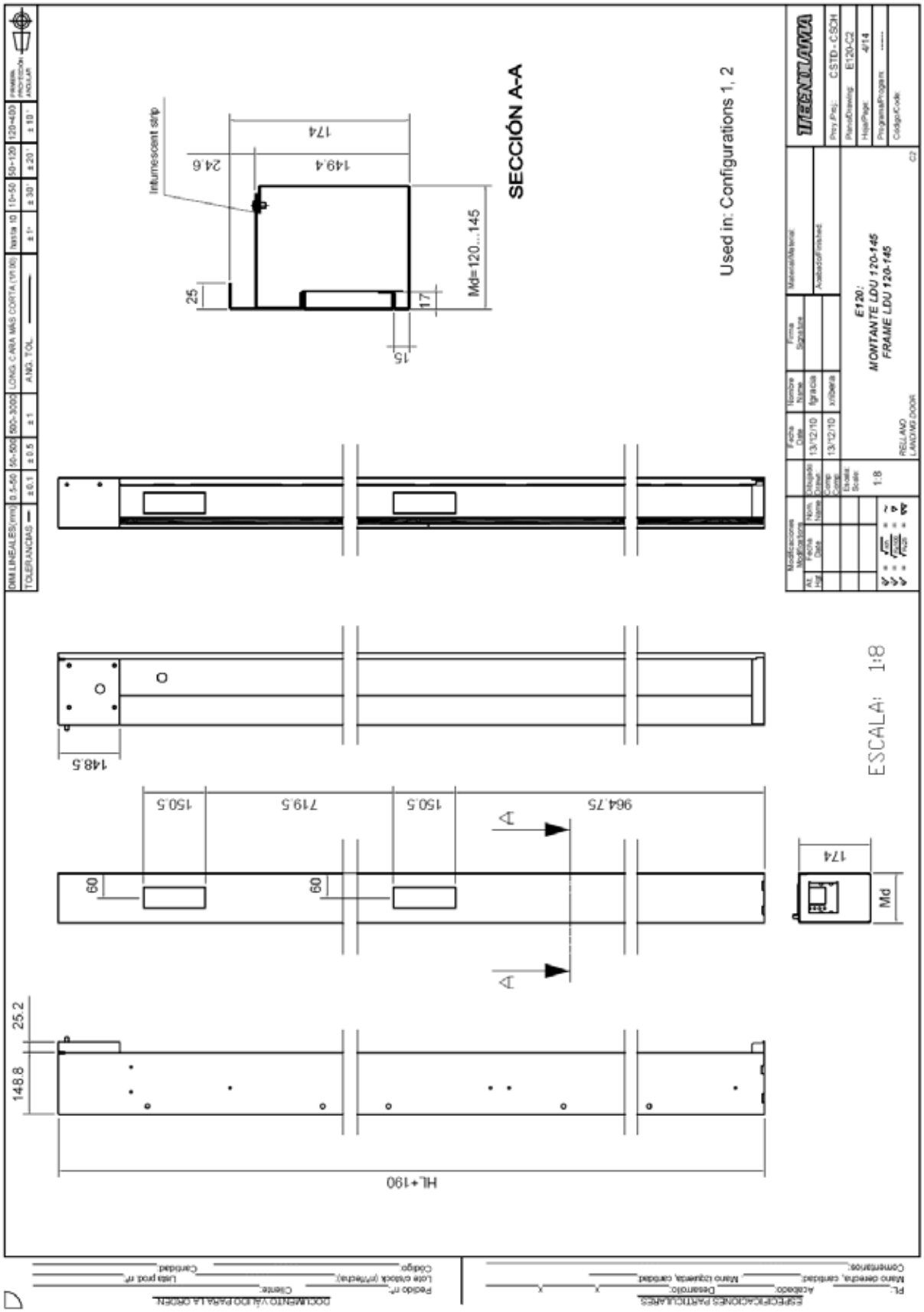
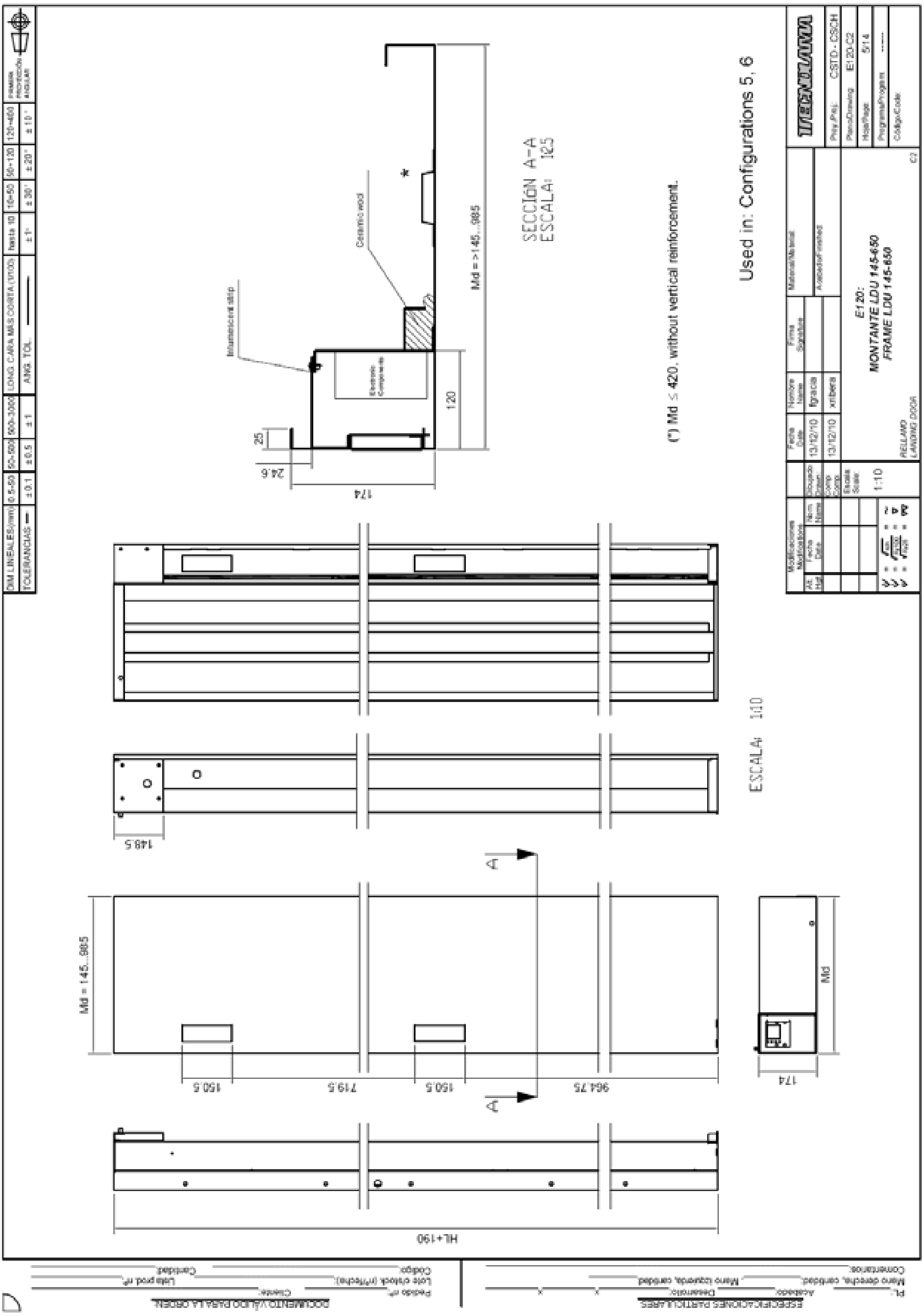
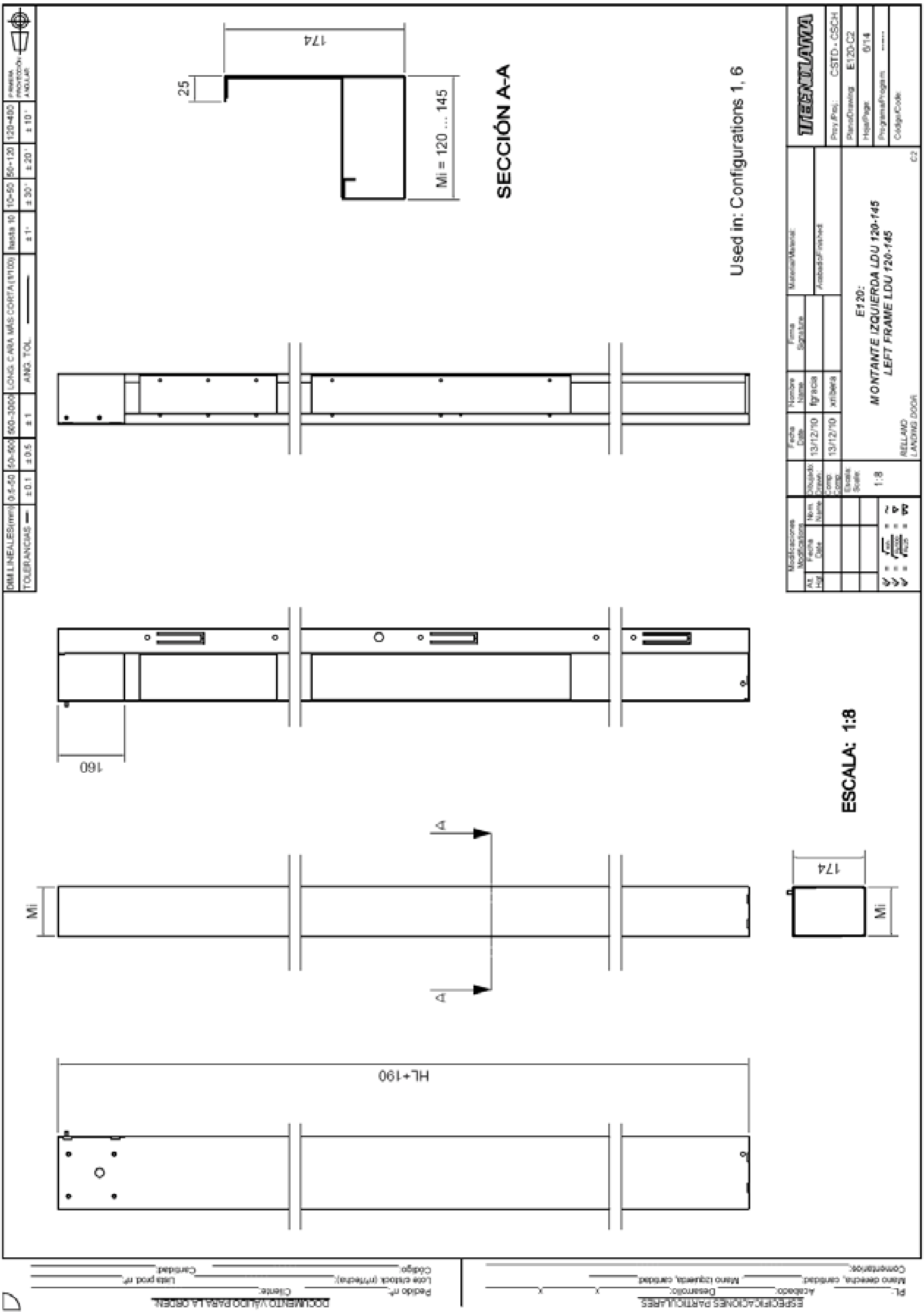
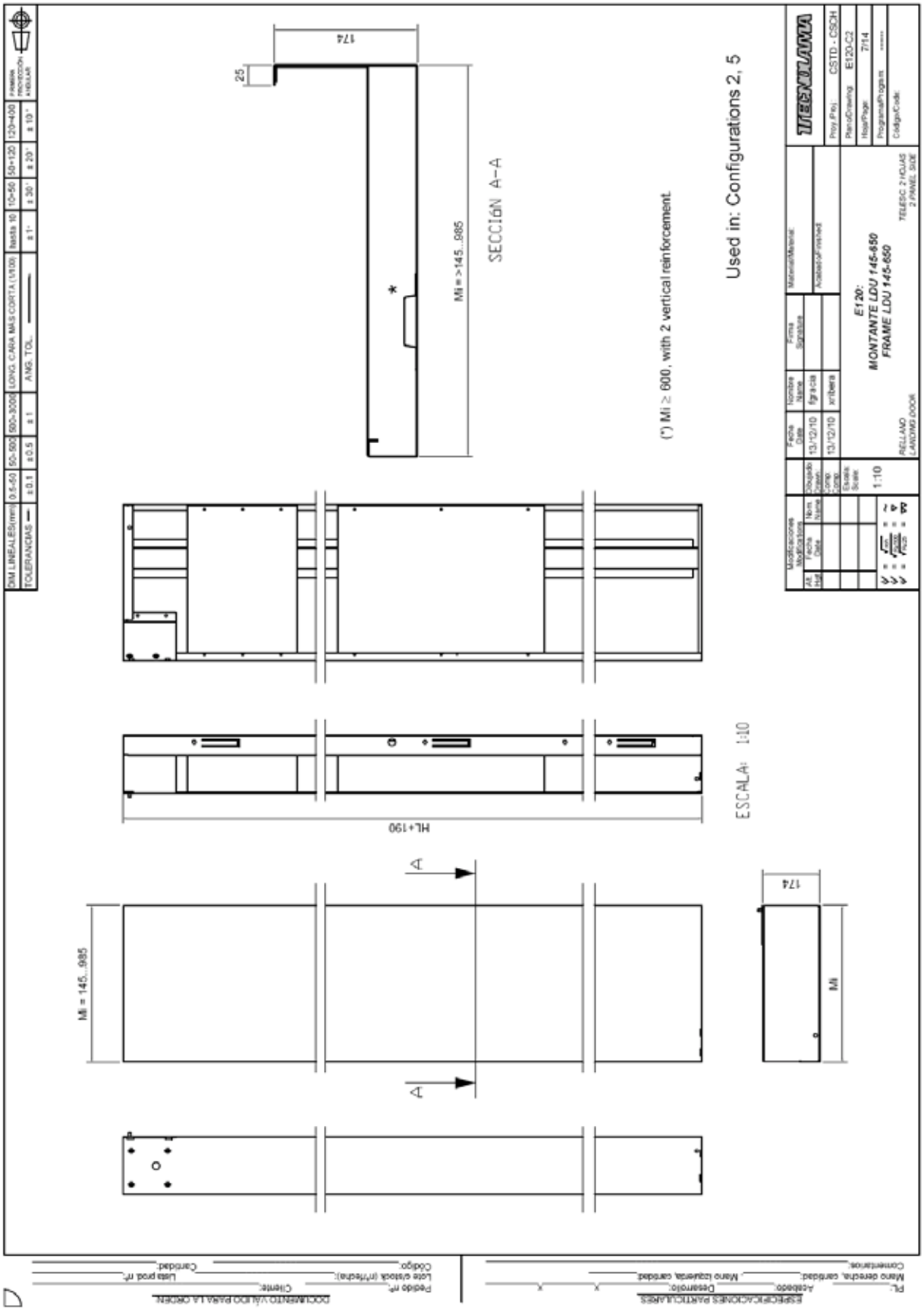

TOTAL HEIGHT = 2190...2490
CLEAR HEIGHT (HL) = 2000...2300
160
20...160
60-174
113 or 227
Behind wall
Concrete filled
10
35...75
Without adjustment
With adjustment
(*) Minim 20 to all covered.
>145...985
600...1100
840...3070
890...3070
Min.20
120...145
1390
E120:
INSTALACIONES / INSTALLATIONS
PELLANO
LANDING DOOR
C2
CSTD-CSCH
E120-C2
2/14

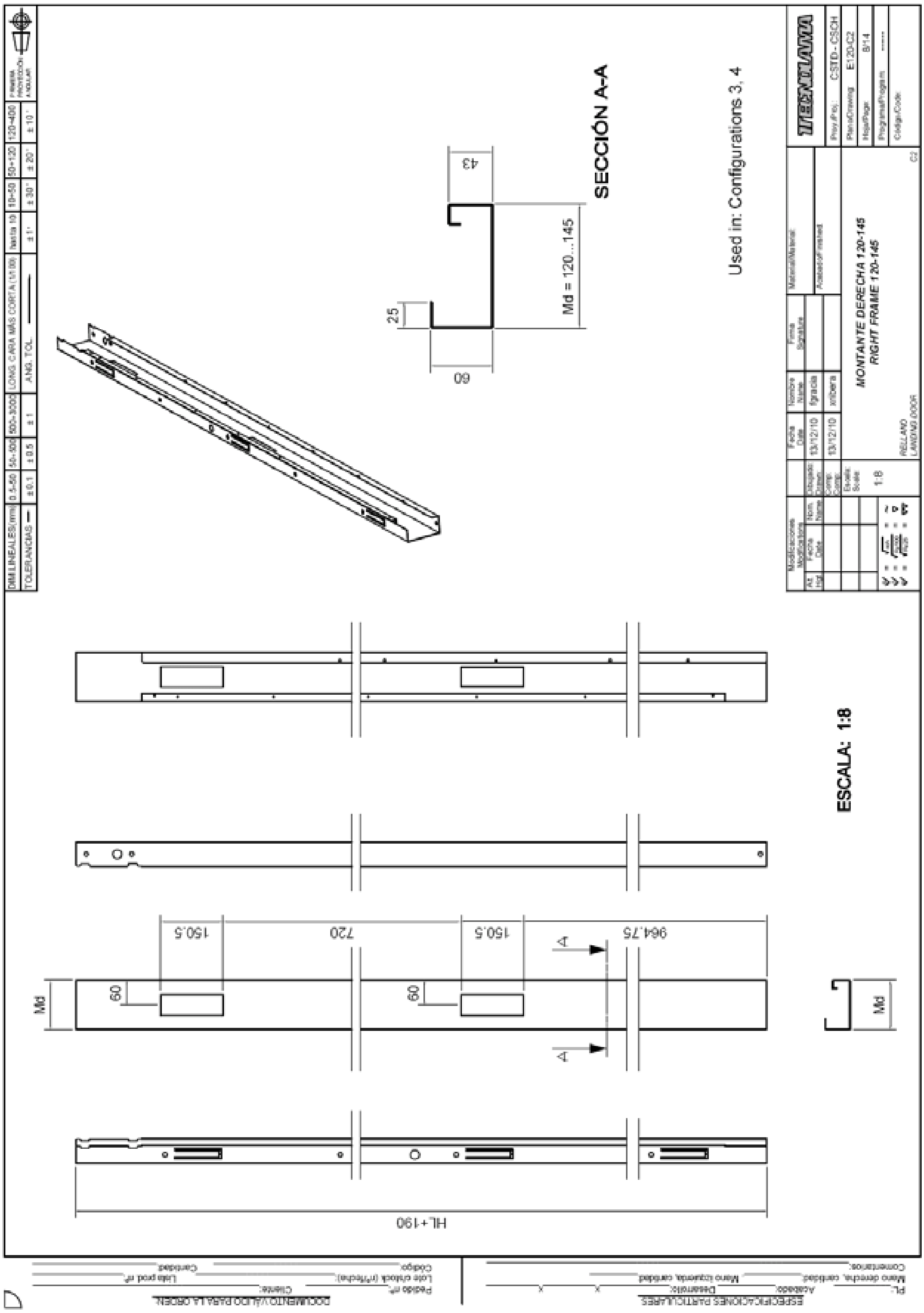

DETALLE "C"
DETALLE "D"
DETALLE "B"
DETALLE "A"
TOTAL HEIGHT = 2190...2490
CLEAR HEIGHT (HL) = 2000...2300
CLEAR HEIGHT = 2000...2300
840...2400
600...1100
840...3070
>145...985
ML: Left front, defined by configurations (dwg 6, 7, 10, 11).
MR: Right front, defined by configurations (dwg 4, 5, 8, 9).
PL: Left panel (dwg 12).
PR: Right panel (dwg 12).
H: Header (dwg 13).
AD: Adjustments (dwg 14).
E120:
VISTAS / VIEWS
RELLANO
LANDING DOOR

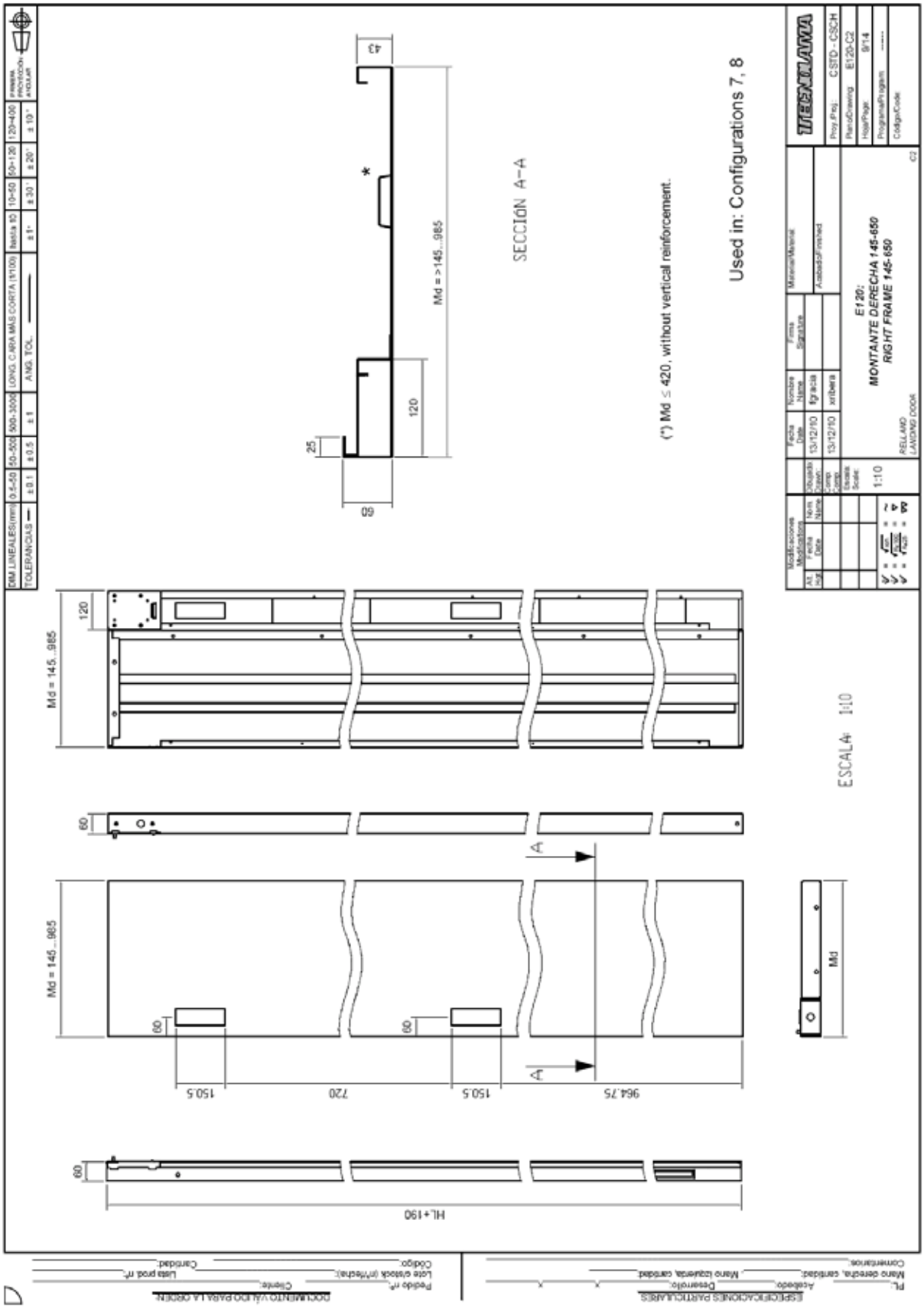

SECCIÓN A-A
Intumescent strip
174
149.4
24.6
25
17
15
Md=120...145
Used in: Configurations 1, 2
148.5
150.5
719.5
150.5
964.75
60
60
A
A
174
Md
25.2
148.8
HL+190
ESCALA: 1:8
E120:
MONTANTE LDU 120-145
FRAME LDU 120-145
REILLANO
LANDING DOOR
TRENTAMA
Proy./Proj.: CSTD-CSCH
Plano/Drawing: E120-C2
Hoja/Page: 4/14
Programa/Program: ----
Código/Code:
Materiales/Material:
Acabado/Finished:
Nombre Name: Ignacio
Iribarra
Fecha Date: 13/12/10
13/12/10
Escala Scale: 1:8
DIM LINEALES(mm) 0.5-50 50-500 500-3000 LONG. CARA MÁS CORTA (m/m) hasta 10 10-50 50-120 120-400 mayor
TOLERANCIAS ±0.1 ±0.5 ±1 ANG. TOL. ±1° ±30' ±20' ±10'
DOCUMENTO VÁLIDO PARA LA ORDEN
ESPECIFICACIONES PARTICULARES

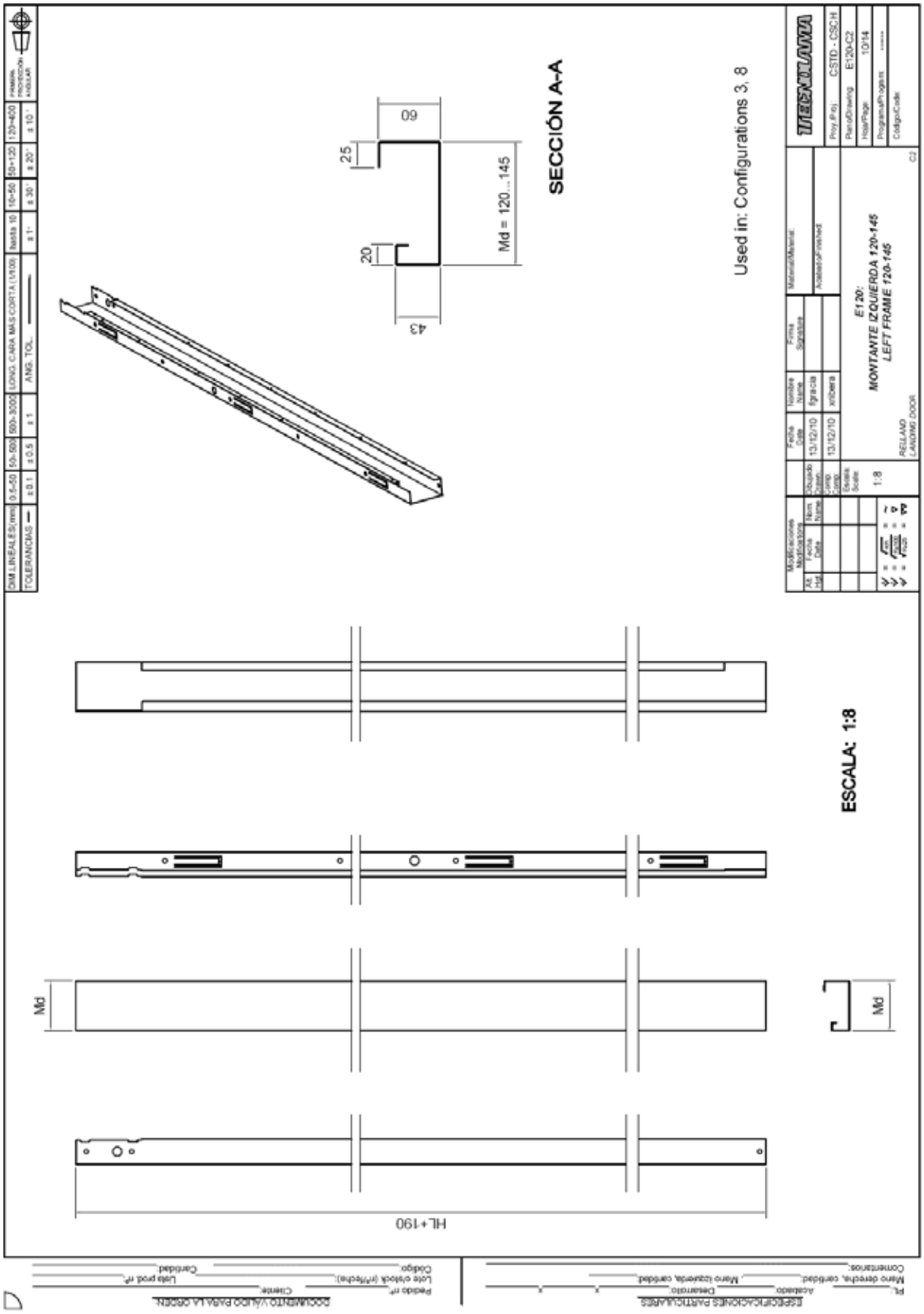

Used in: Configurations 5, 6
(*) Md ≤ 420, without vertical reinforcement.
SECCIÓN A-A
ESCALA: 1:2.5
Ceramic wool
Intumescent strip
Md = >145...985
ESCALA: 1:10
HL+190
Md = 145...985
E120:
MONTANTE LDU 145-650
FRAME LDU 145-650
RELLANO
LANDING DOOR
CSTD - CSCH
E120-C2
5/14
fgracia
xnibera
13/12/10
1:10

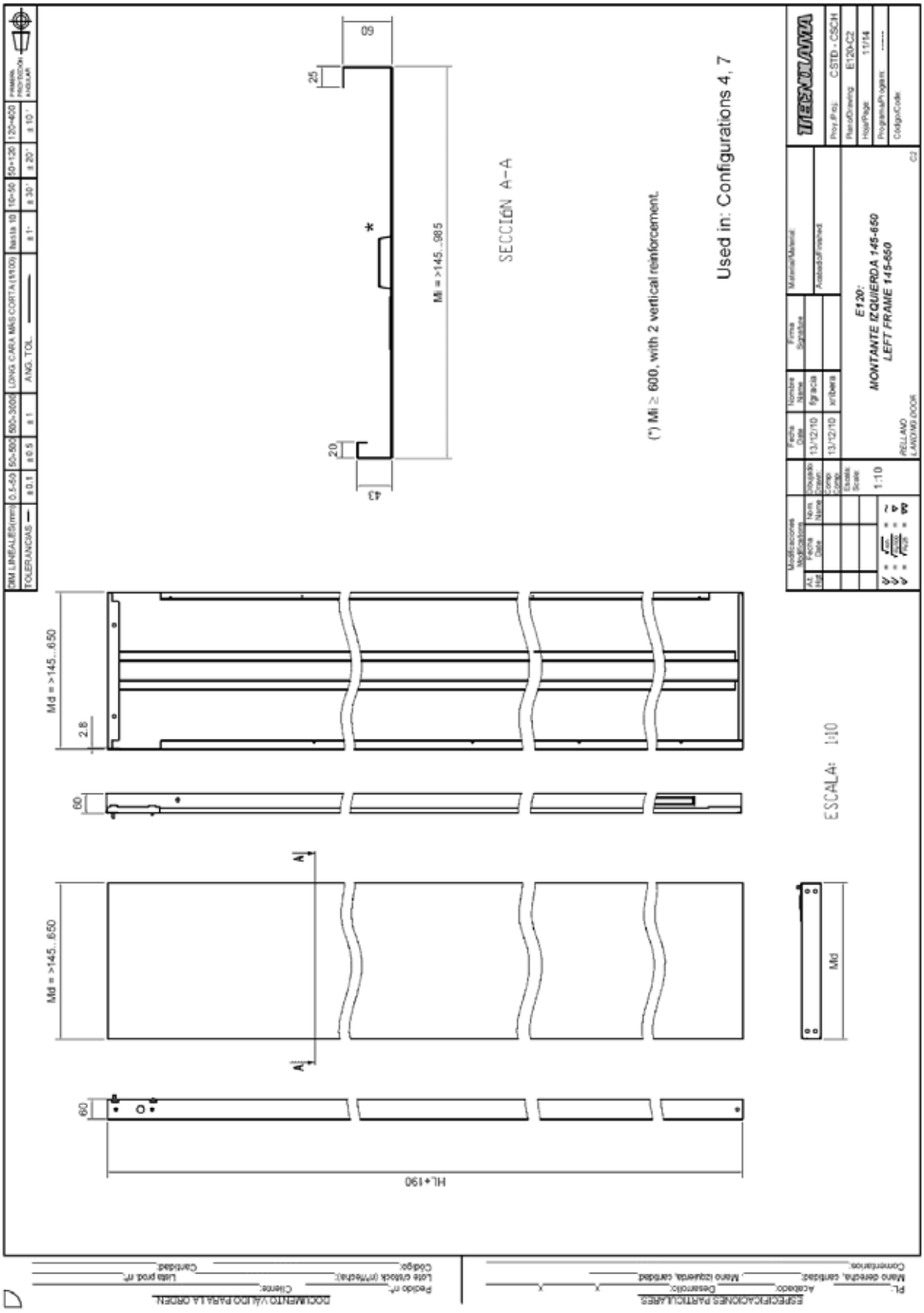

SECCIÓN A-A
174
25
Mi = 120 ... 145
Used in: Configurations 1, 6
ESCALA: 1:8
160
HL+190
Mi
174
E120:
MONTANTE IZQUIERDA LDU 120-145
LEFT FRAME LDU 120-145
RELLANO
LANDING DOOR
1:8
E120-C2
6/14
CSTD - CSCH
13/12/10
fgracia
xribera

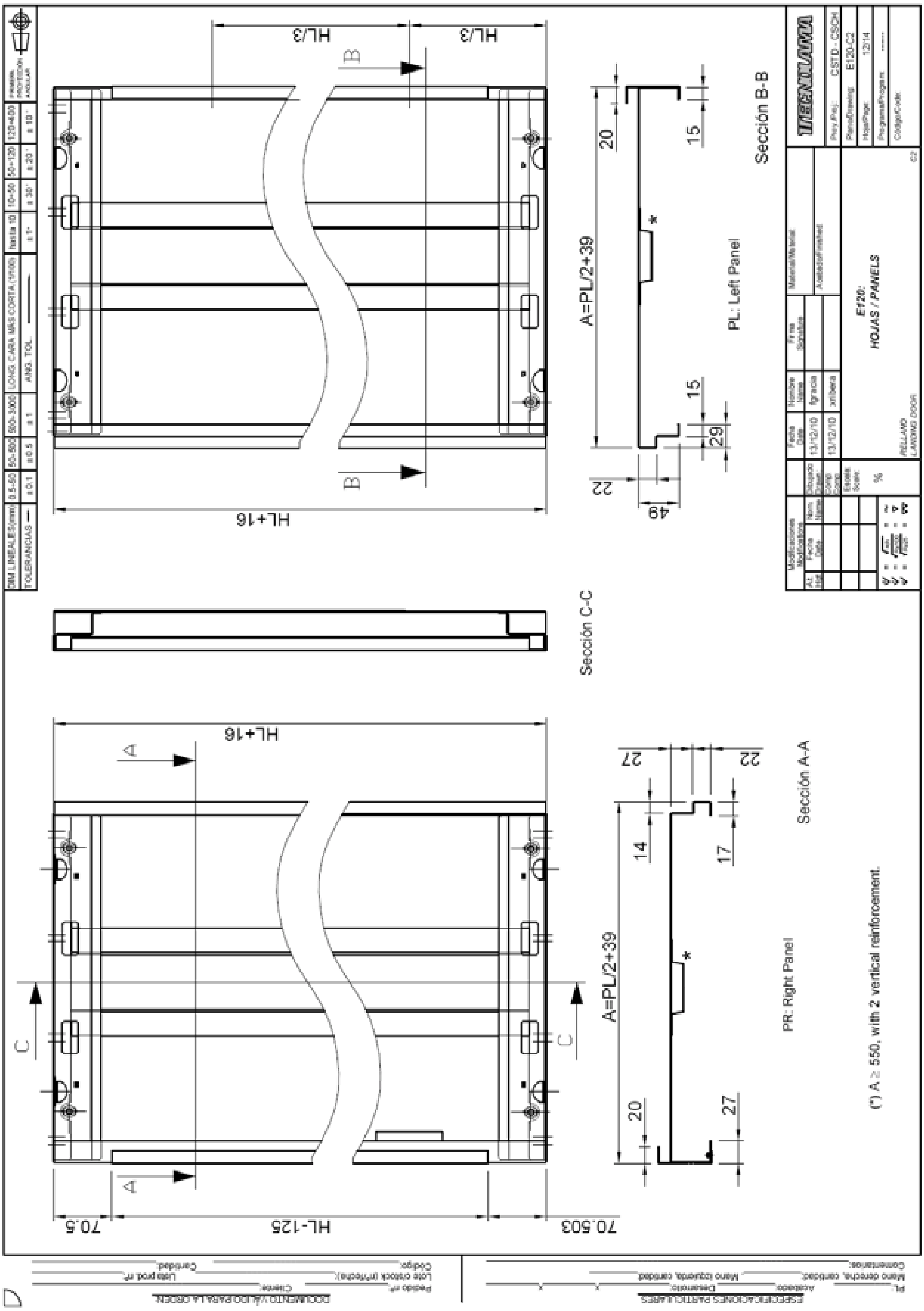

SECCIÓN A-A
174
25
Mi = >145...985
*
(*) Mi ≥ 600, with 2 vertical reinforcement
Used in: Configurations 2, 5
ESCALA: 1:10
HL+190
Mi = 145...985
174
Mi
A
A
E120:
MONTANTE LDU 145-650
FRAME LDU 145-650
1:10

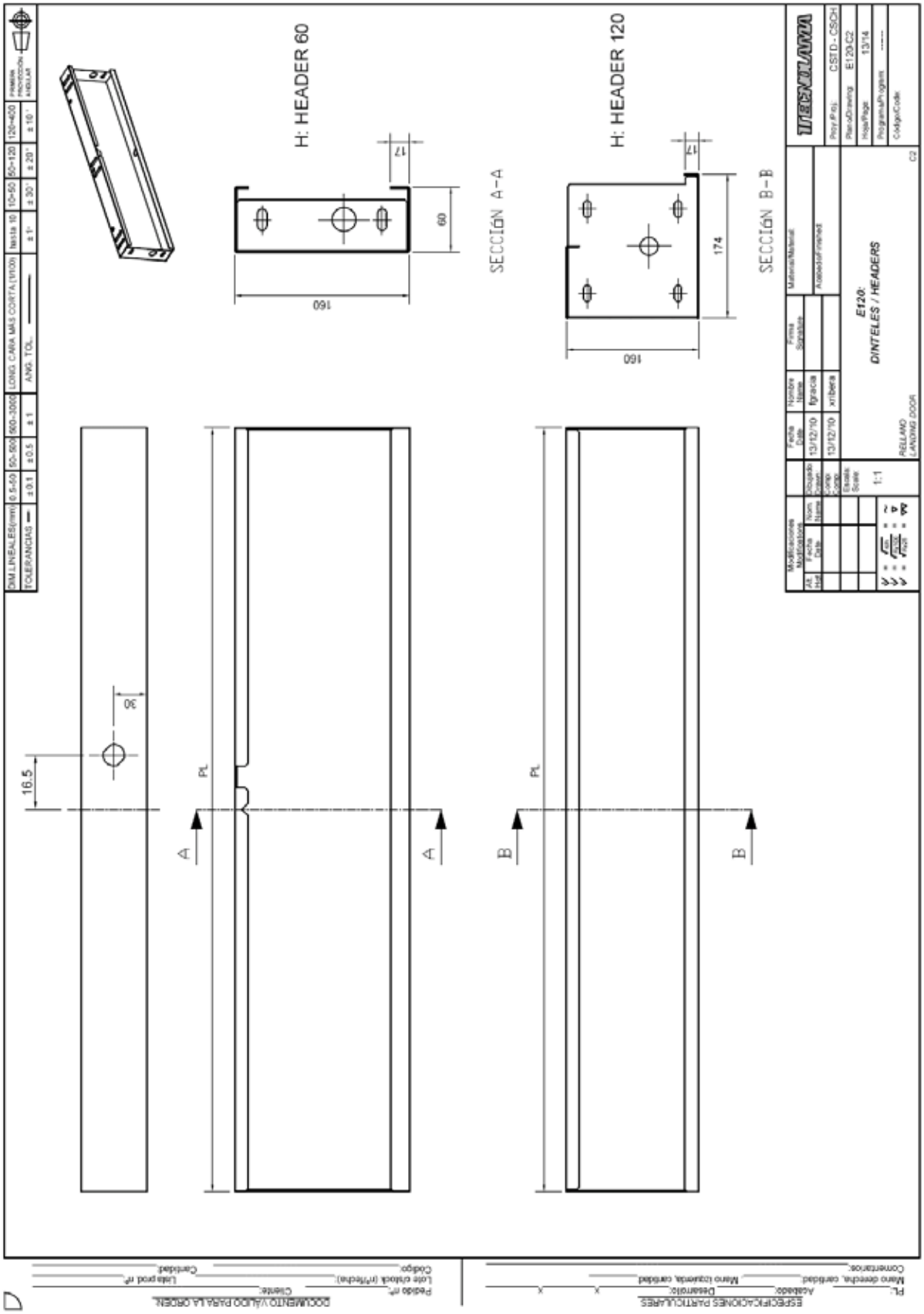

SECCIÓN A-A
Used in: Configurations 3, 4
Md = 120...145
ESCALA: 1:8
MONTANTE DERECHA 120-145
RIGHT FRAME 120-145
964.75
720
150.5
HL+190


SECCIÓN A-A
Md = >145...985
(*) Md ≤ 420, without vertical reinforcement.
Used in: Configurations 7, 8
Md = 145...985
HL+190
964.75
720
150.5
ESCALA: 1:10
E120:
MONTANTE DERECHA 145-650
RIGHT FRAME 145-650
1:10


SECCIÓN A-A
Md = 120 ... 145
60
25
20
43
Used in: Configurations 3, 8
ESCALA: 1:8
HL+190
Md
E120:
MONTANTE IZQUIERDA 120-145
LEFT FRAME 120-145
REILAND
LANDING DOOR
C2
C-STD - CSCH
E120-C2
10/14
TOLERANCIAS
DIM. LINEALES (mm)
LONG. CARA MÁS CORTA (1/100)
ANG. TOL.
Fecha Date
13/12/10
Nombre Name
fgracia
Firma Signature
Material/Material:
Acabado/Finished:
Modificaciones Modifications
Escala Scale
1:8
Proy.Ref.(s)
Plano/Drawing
Hoja/Page
Programa/Program
Código/Code
DESPIECIFICACIONES PARTICULARES
DOCUMENTO VÁLIDO PARA LA ORDEN


SECCIÓN A-A
Used in: Configurations 4, 7
(*) Mi ≥ 600, with 2 vertical reinforcement.
Mi = >145..985
Md = >145..650
HL+190
ESCALA: 1:10
E120:
MONTANTE IZQUIERDA 145-650
LEFT FRAME 145-650
C2

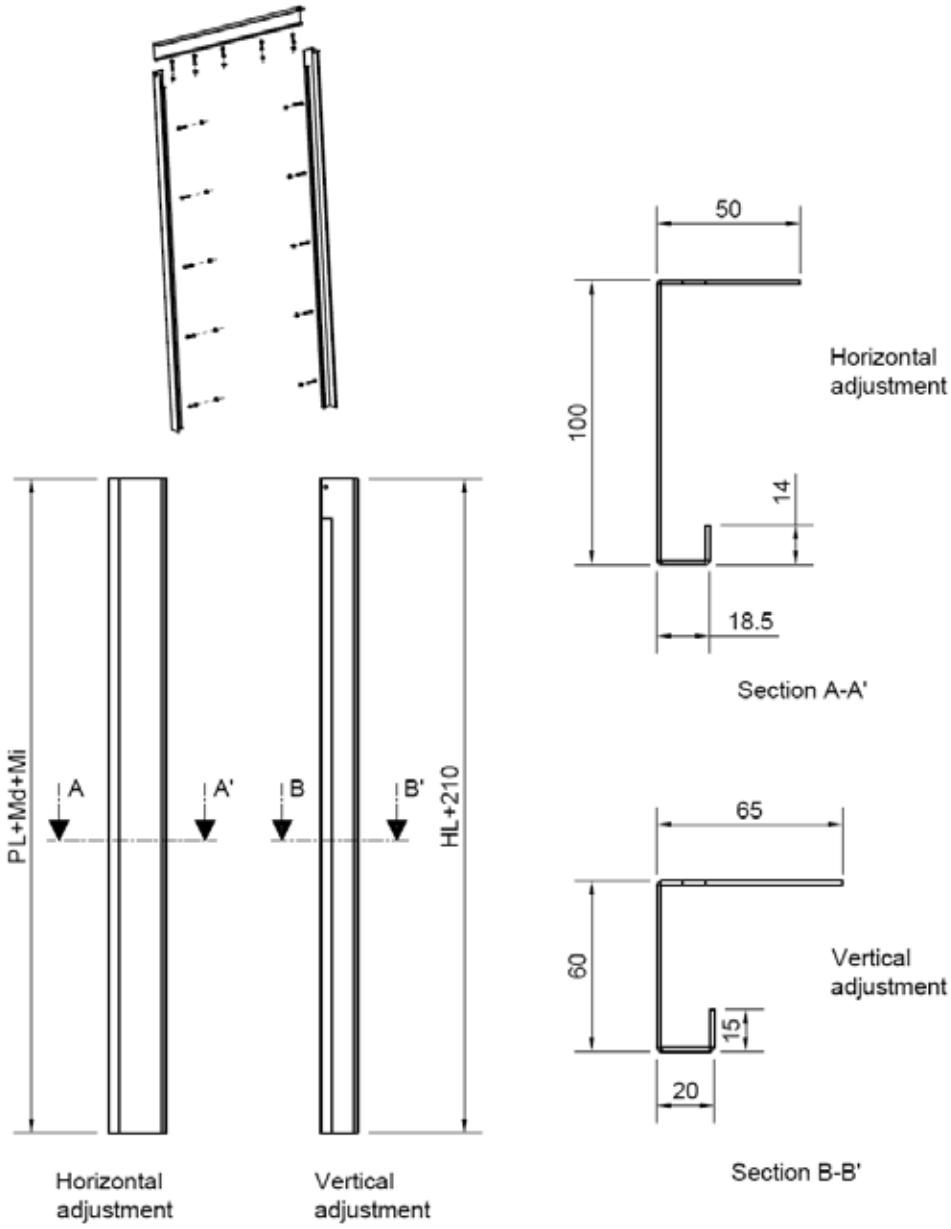

HL/3
HL/3
HL+16
A=PL/2+39
PL: Left Panel
Sección B-B
Sección C-C
HL+16
A=PL/2+39
PR: Right Panel
Sección A-A
(*) A ≥ 550, with 2 vertical reinforcement.
70.5
HL-125
70.503
E120:
HOJAS / PANELS


H: HEADER 60
H: HEADER 120
SECCIÓN A-A
SECCIÓN B-B
160
60
17
174
30
16.5
PL
A
B
E120:
DINTELES / HEADERS
REILLANO
LANDING DOOR
CSTD - CSCH
E120.C2
13/14
C2
1:1
13/12/10
fgarcia
xrribera
DIM LINEALES(mm)
TOLERANCIAS
LONG. CARA MAS CORTA (mm)
ANG. TOL.
DOCUMENTO VÁLIDO PARA LA ORDEN
ESPECIFICACIONES PARTICULARES

DIM.LINEALES(mm)	0.5÷50	50÷500	500÷3000	LONG. CARA MÁS CORTA (1/100)	hasta 10	10÷50	50÷120	120÷400	PRIMERA PROYECCIÓN ANGULAR
TOLERANCIAS —	± 0.1	± 0.5	± 1	ANG. TOL. ———	± 1°	± 30 '	± 20 '	± 10 '	

DOCUMENTO VÁLIDO PARA LA ORDEN:
Pedido nº: Cliente: Lista prod. nº:
Lote c/stock (nº/fecha): Cantidad:
Código:

ESPECIFICACIONES PARTICULARES:
Acabado: Desarrollo: x
Mano derecha, cantidad: Mano izquierda, cantidad: x
PL: Comentarios:

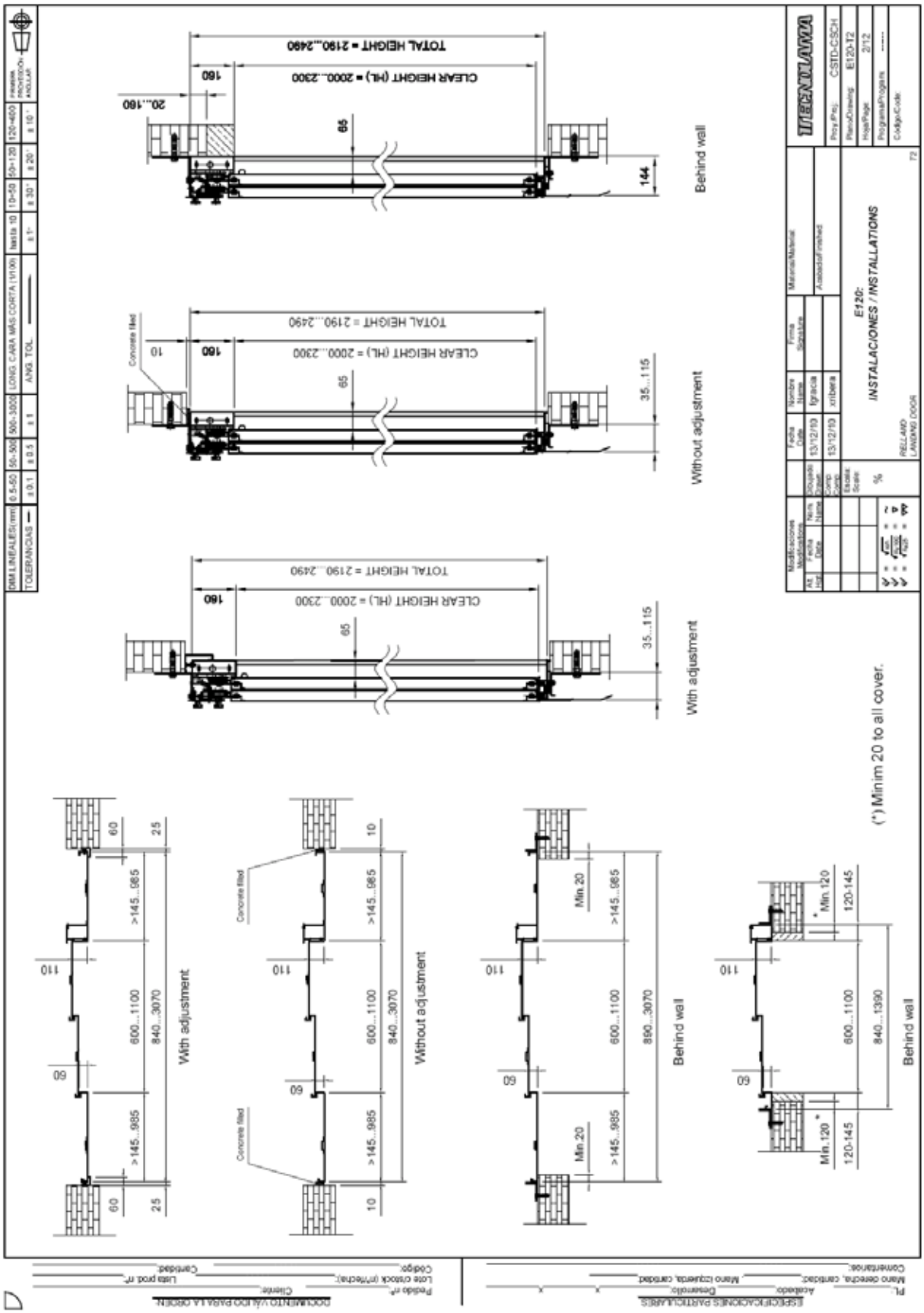


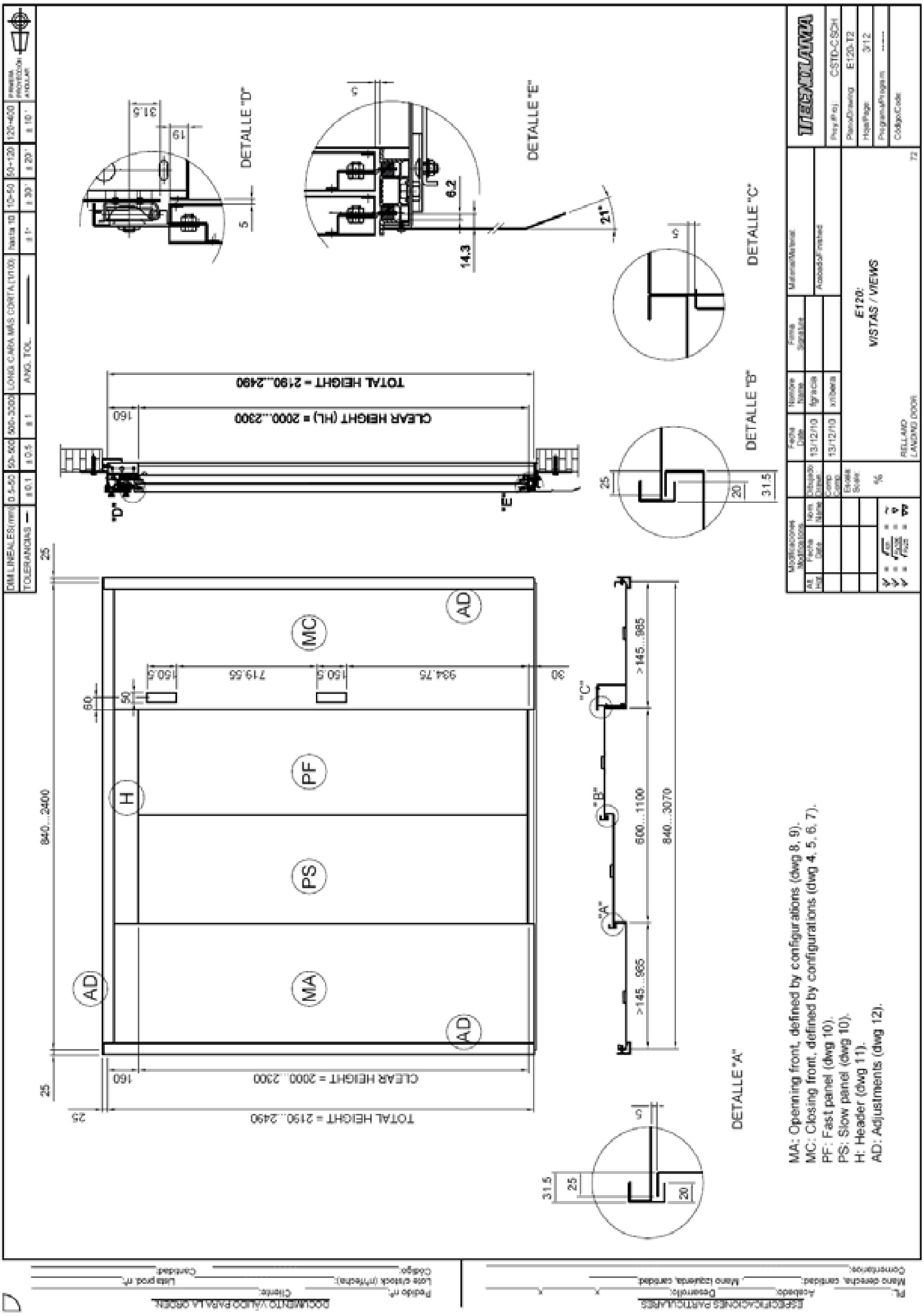
Modificaciones Modifications				Fecha Date	Nombre Name	Firma Signature	Material/Material:	TECNOLAMA
Alt. Hgt.	Fecha Date	Nom. Name	Dibujado Drawn:	13/12/10	fgracia		Acabado/Finished:	
			Comp: Comp:	13/12/10	xribera			Proy./Proj.: CSTD - CSCH
			Escala: Scale: %					Plano/Drawing: E120-C2
								Hoja/Page: 14/14
								Programa/Program: ------
								Código/Code:

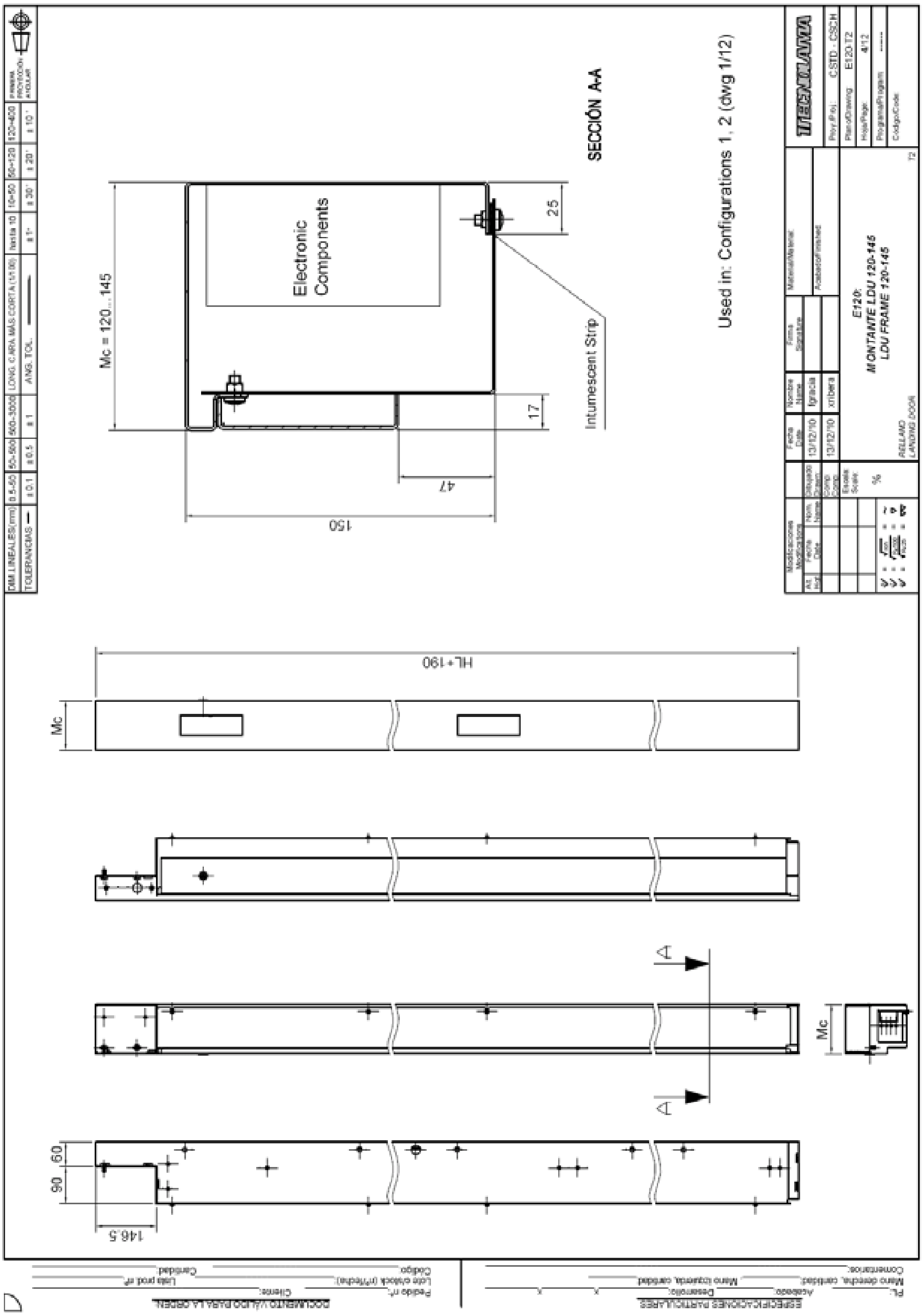
E120: TAPAJUNTAS ADJUSTMENTS

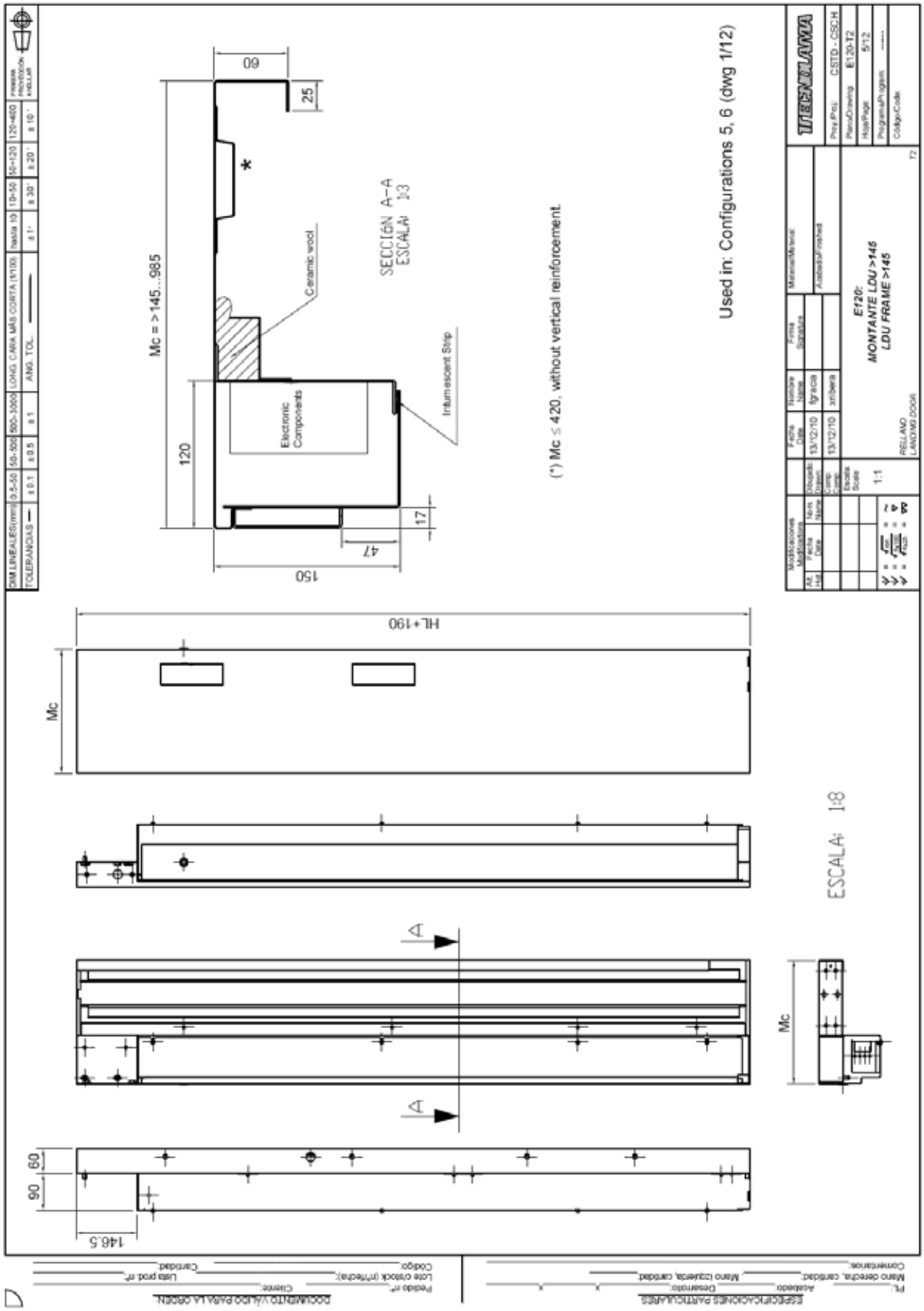
RELLANO
LANDING DOOR

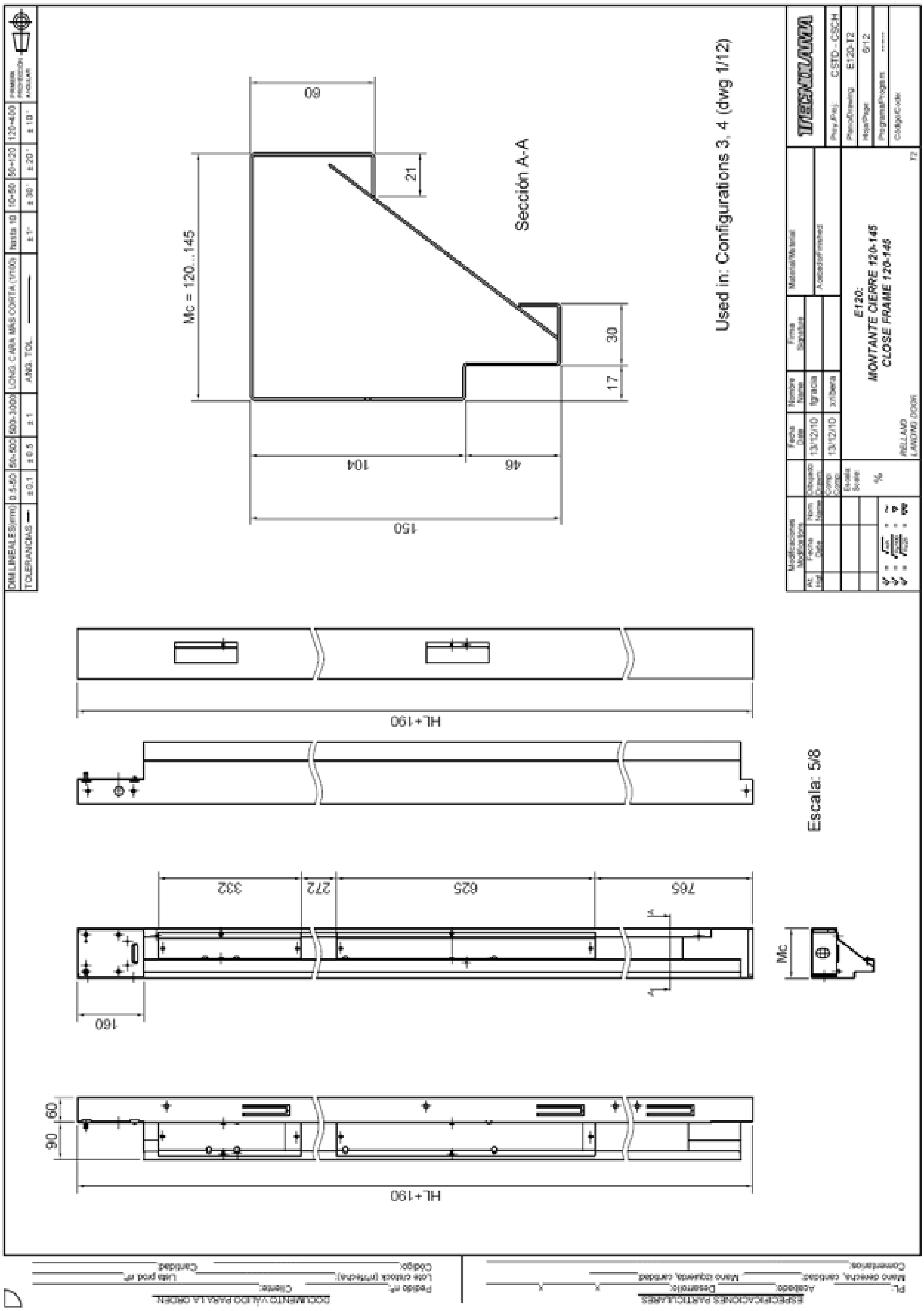
C2

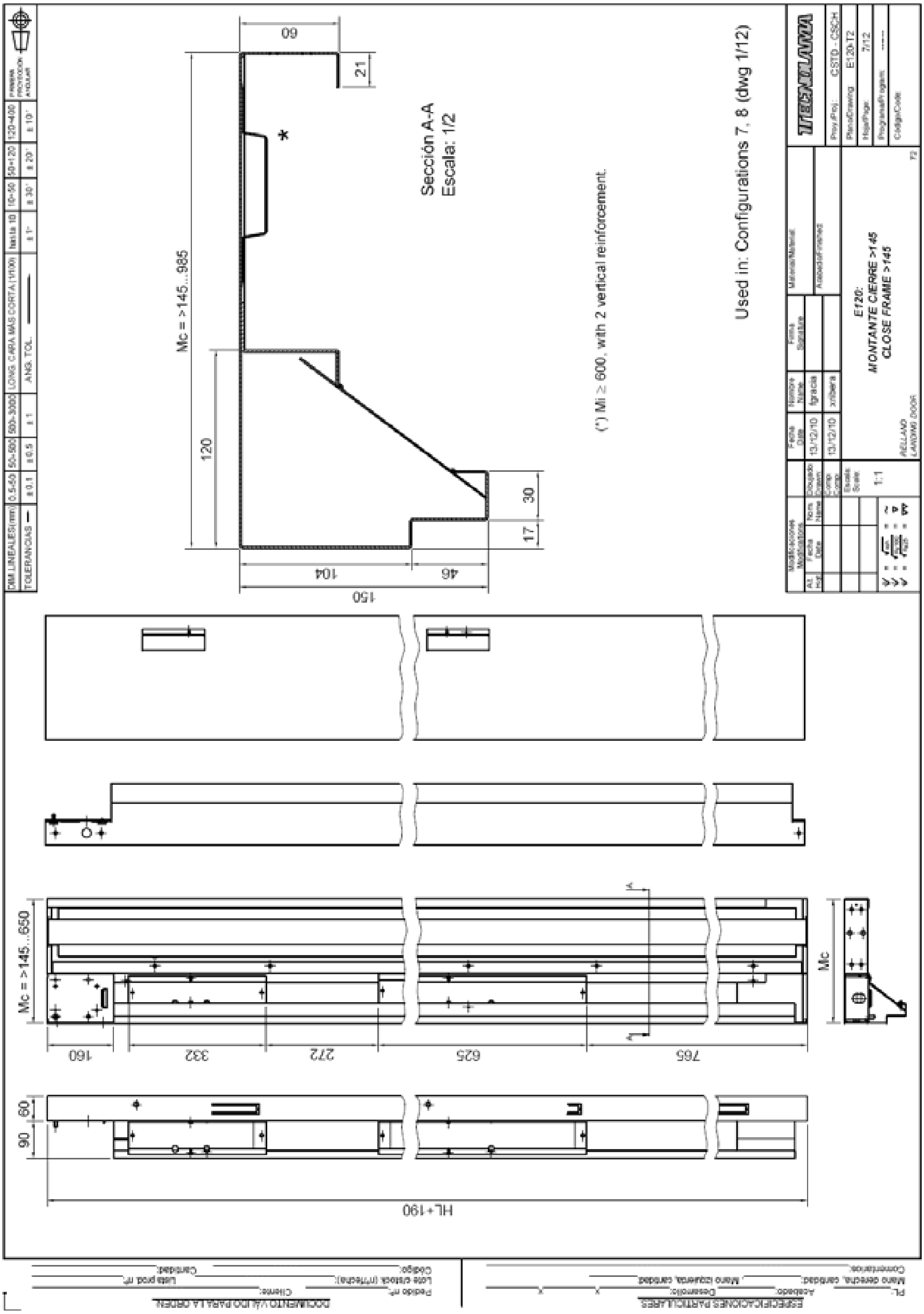

TOTAL HEIGHT = 2190...2490
CLEAR HEIGHT (HL) = 2000...2300
160
65
144
20...160
Behind wall
Concrete filled
10
35...115
Without adjustment
With adjustment
>145...985
600...1100
840...3070
890...3070
1390
Min.20
Min.120 *
120-145
110
60
25
(*) Minim 20 to all cover.
INSTALACIONES / INSTALLATIONS
E120:
CSTD-CSCH
E120-12
2/12
13/12/10
tgracia
criberia
RELLAND
LANDING DOOR
72

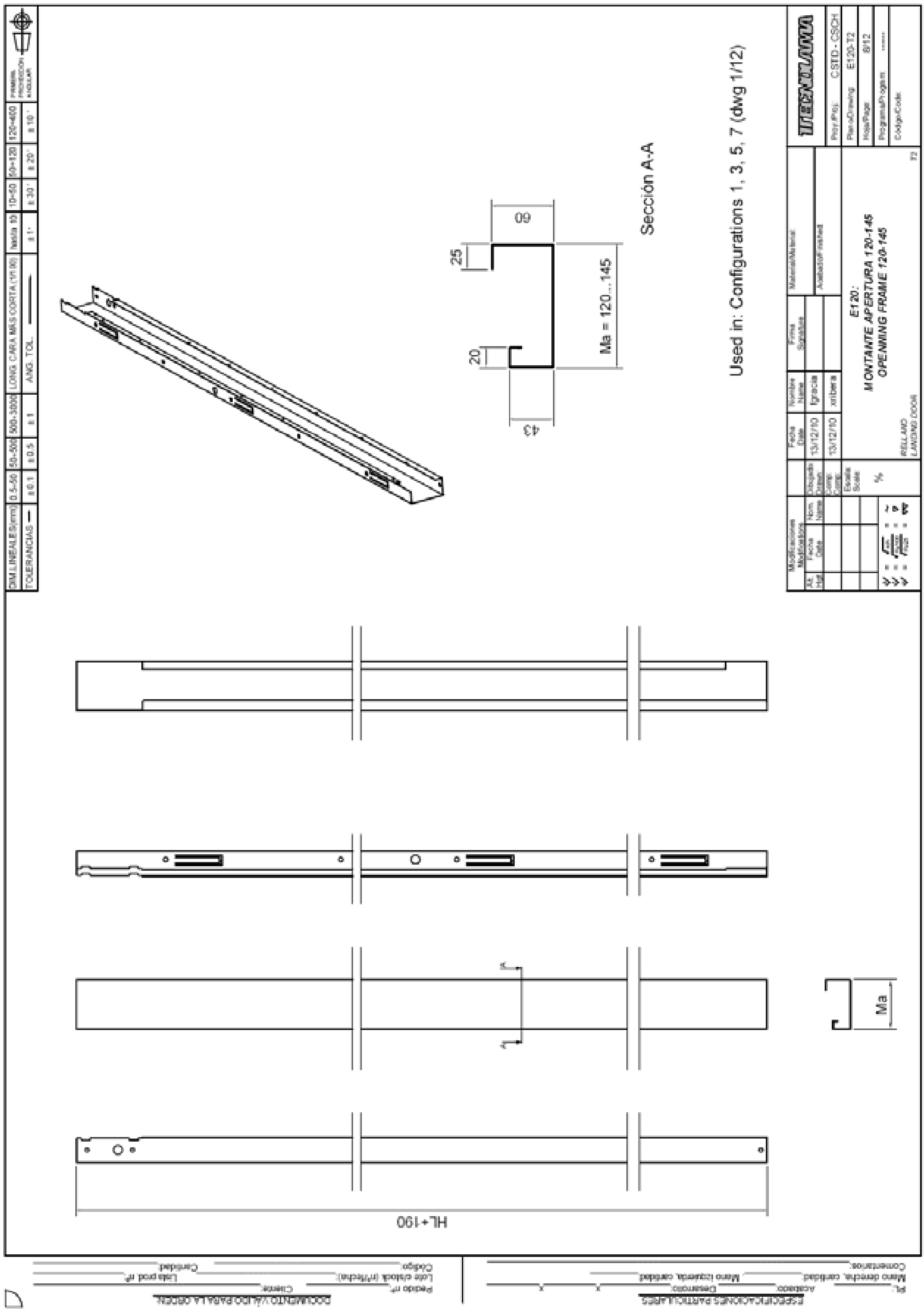

DETALLE "D"
DETALLE "E"
DETALLE "C"
DETALLE "B"
DETALLE "A"
TOTAL HEIGHT = 2190...2490
CLEAR HEIGHT (HL) = 2000...2300
840...2400
840...3070
600...1100
>145...965
E120:
VISTAS / VIEWS
RELLANO
LANDING DOOR
C-STD-CSCH
E120-T2
3/12
MA: Openning front, defined by configurations (dwg 8, 9).
MC: Closing front, defined by configurations (dwg 4, 5, 6, 7).
PF: Fast panel (dwg 10).
PS: Slow panel (dwg 10).
H: Header (dwg 11).
AD: Adjustments (dwg 12).

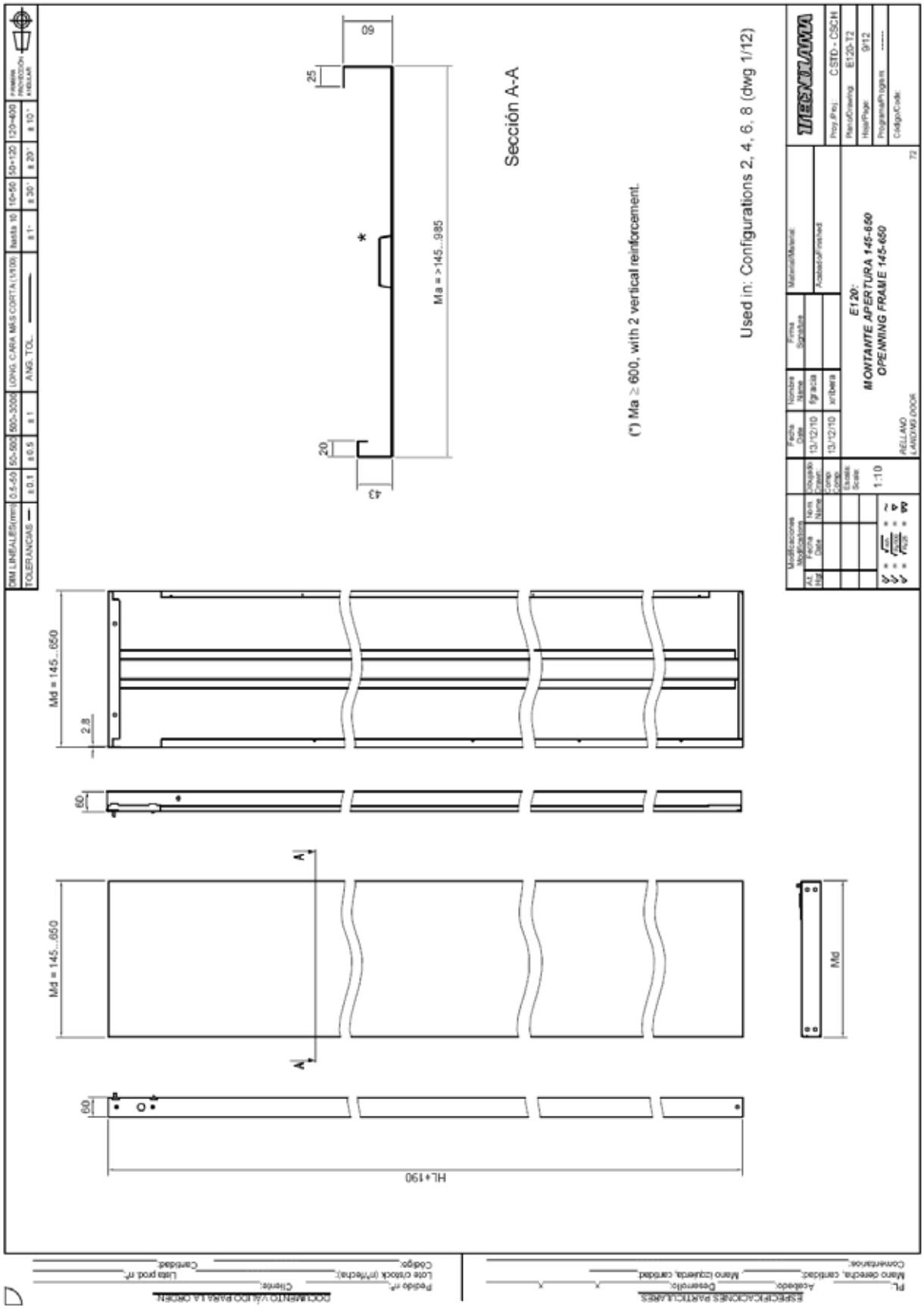

SECCIÓN A-A
Used in: Configurations 1, 2 (dwg 1/12)
Electronic Components
Intumescent Strip
Mc = 120 ... 145
25
17
47
150
HL+190
Mc
90
60
146.5
A
E120:
MONTANTE LDU 120-145
LDU FRAME 120-145
RELLANO
LANDING DOOR
CSTD - CSCH
E120-T2
4/12
Igracia
xribera
13/12/10

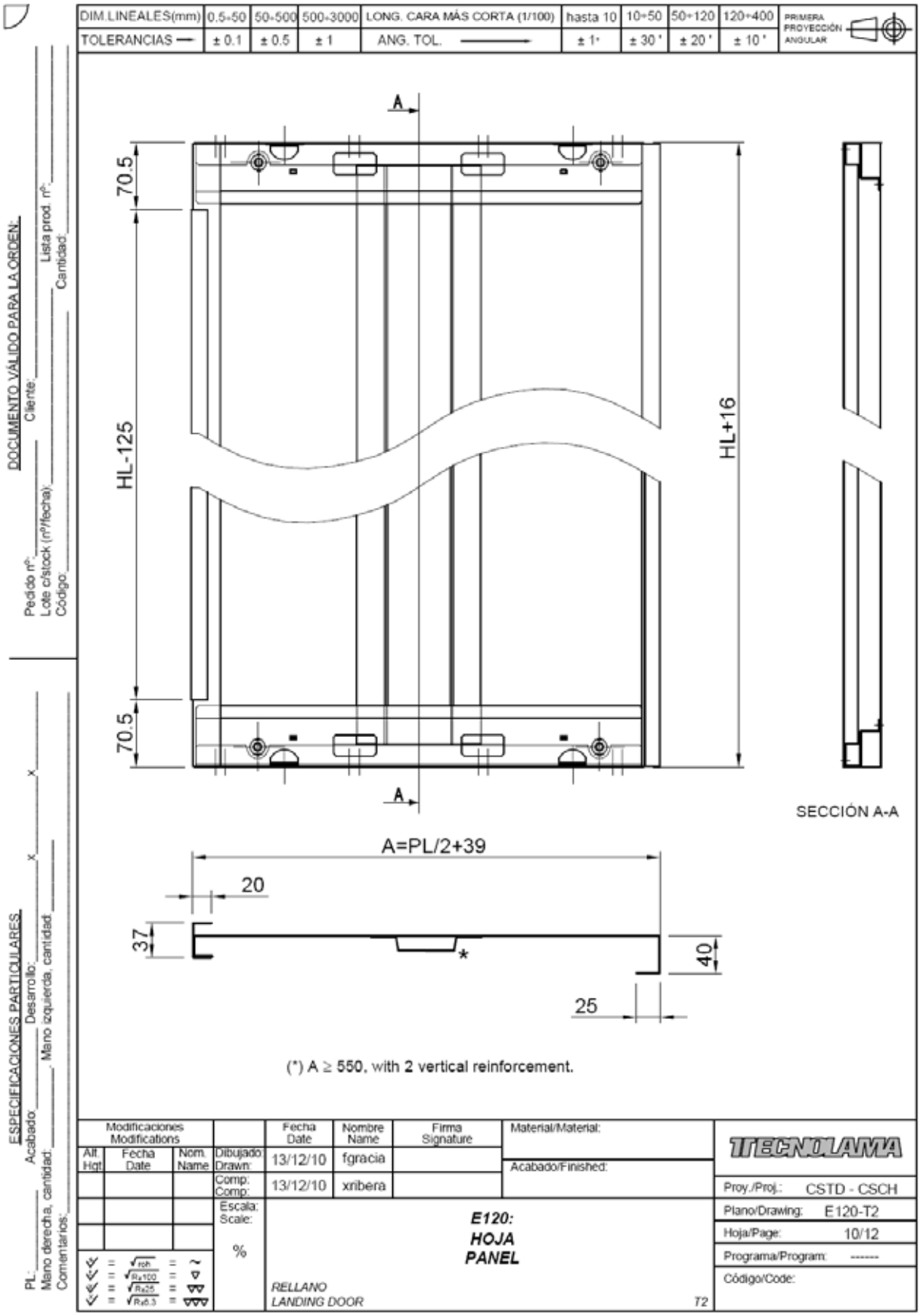

Mc = >145...985
60
25
*
Ceramic wool
SECCIÓN A-A
ESCALA 1:3
Intumescent Strip
Electronic Components
120
17
47
150
(*) Mc ≤ 420, without vertical reinforcement.
Used in: Configurations 5, 6 (dwg 1/12)
THERMAMA
E120:
MONTANTE LDU >145
LDU FRAME >145
HL+190
Mc
ESCALA: 1:8
A
A
60
90
146.5
Mc

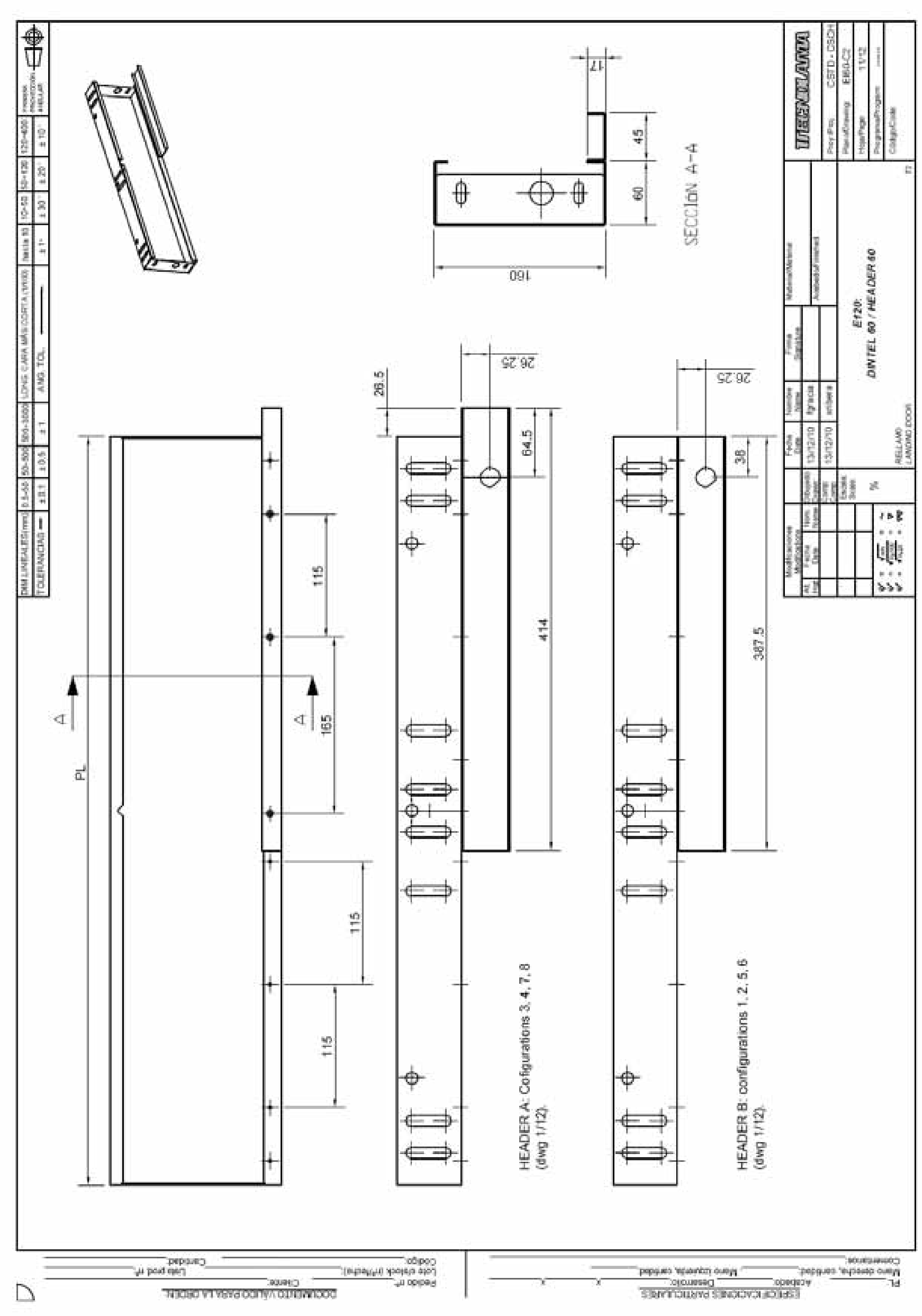

Sección A-A
Used in: Configurations 3, 4 (dwg 1/12)
Mc = 120...145
Escala: 5/8
HL+190
E120:
MONTANTE CIERRE 120-145
CLOSE FRAME 120-145
RELLANO
LANDING DOOR


Sección A-A
Escala: 1/2
Used in: Configurations 7, 8 (dwg 1/12)
(*) Mi ≥ 600, with 2 vertical reinforcement.
Mc = >145...985
Mc = >145...650
HL+190
E120:
MONTANTE CIERRE >145
CLOSE FRAME >145


Used in: Configurations 1, 3, 5, 7 (dwg 1/12)
Sección A-A
Ma = 120...145
60
25
20
43
HL+190
Ma
E120:
MONTANTE APERTURA 120-145
OPENING FRAME 120-145
RELLANO
LANDING DOOR
CSTD - CSCH
E120-T2
8/12
13/12/10
lgracia
xribera
DOCUMENTO VÁLIDO PARA LA ORDEN
ESPECIFICACIONES PARTICULARES


Sección A-A
Ma = >145...985
(*) Ma ≥ 600, with 2 vertical reinforcement.
Used in: Configurations 2, 4, 6, 8 (dwg 1/12)
Md = 145...650
HL+190
MONTANTE APERTURA 145-650
OPENWING FRAME 145-650
E120:
1:10

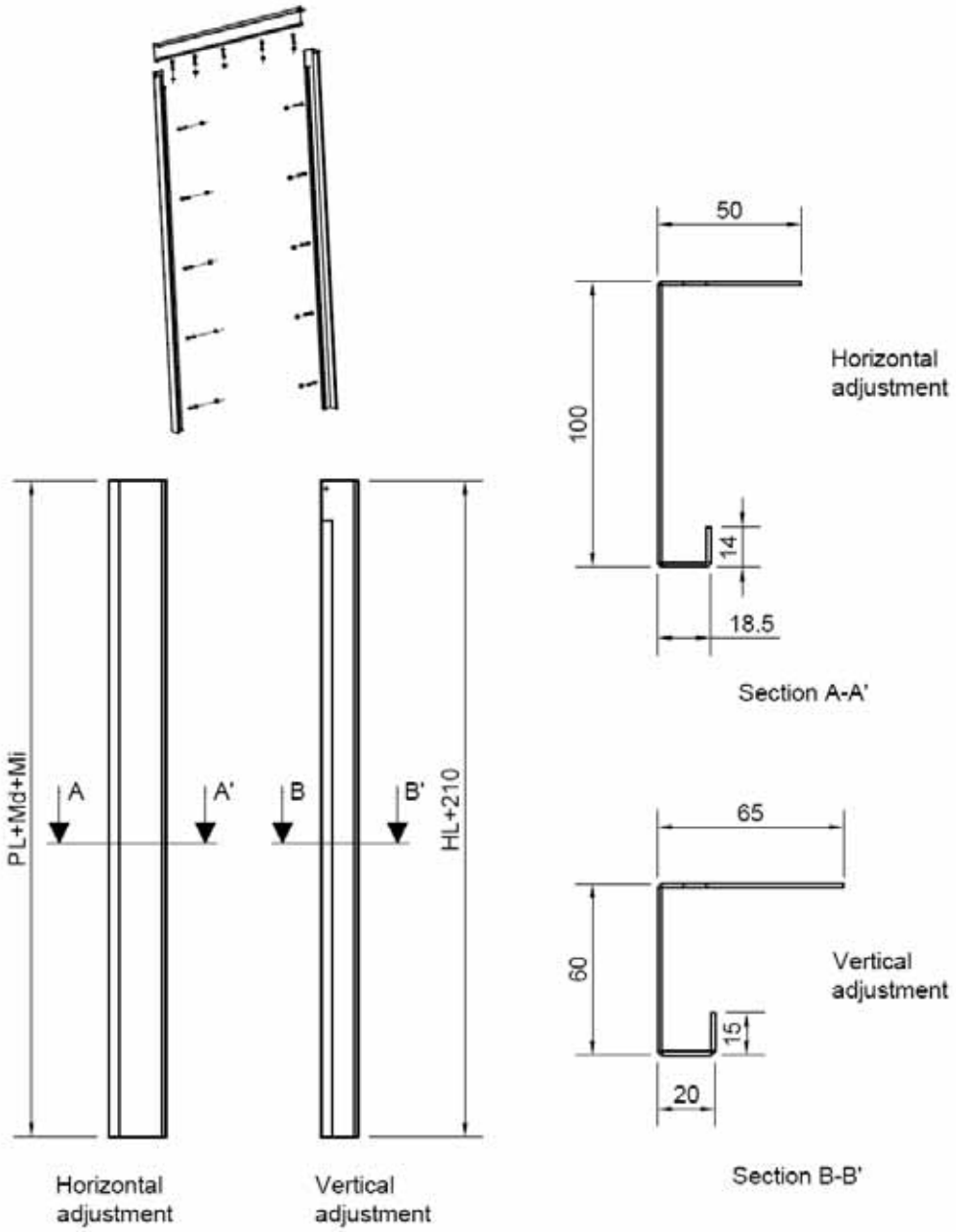

DIM.LINEALES(mm) 0.5÷50 50÷500 500÷3000 LONG. CARA MÁS CORTA (1/100) hasta 10 10÷50 50÷120 120÷400
TOLERANCIAS ± 0.1 ± 0.5 ± 1 ANG. TOL. ± 1° ± 30' ± 20' ± 10'
PRIMERA PROYECCIÓN ANGULAR
A
70.5
HL-125
70.5
HL+16
SECCIÓN A-A
A=PL/2+39
20
37
40
25
*
(*) A ≥ 550, with 2 vertical reinforcement.
DOCUMENTO VÁLIDO PARA LA ORDEN:
Pedido nº:
Lote c/stock (nº/fecha):
Código:
Cliente:
Lista prod. nº:
Cantidad:
ESPECIFICACIONES PARTICULARES:
Acabado:
Desarrollo:
PL:
Mano derecha, cantidad:
Mano izquierda, cantidad:
Comentarios:
Modificaciones Modifications
Alt. Hgt
Fecha Date
Nom. Name
Dibujado Drawn: 13/12/10 fgracia
Comp. Comp: 13/12/10 xribera
Escala: Scale: %
Fecha Date
Nombre Name
Firma Signature
Material/Material:
Acabado/Finished:
E120:
HOJA
PANEL
RELLANO
LANDING DOOR
T2
TECNOLAMA
Proy./Proj.: CSTD - CSCH
Plano/Drawing: E120-T2
Hoja/Page: 10/12
Programa/Program: ------
Código/Code:


HEADER A: Cofigurations 3, 4, 7, 8
(dwg 1/12).
HEADER B: configurations 1, 2, 5, 6
(dwg 1/12).
SECCIÓN A-A
EI120:
DINTEL 60 / HEADER 60
PL
414
387.5
115
165
160
60
45
17
26.5
26.25
64.5
38

DIM.LINEALES(mm)	0.5-50	50-500	500-3000	LONG. CARA MÁS CORTA (1/100)	hasta 10	10÷50	50÷120	120÷400	PRIMERA PROYECCIÓN ANGULAR
TOLERANCIAS	± 0.1	± 0.5	± 1	ANG. TOL.	± 1°	± 30'	± 20'	± 10'	

DOCUMENTO VÁLIDO PARA LA ORDEN

Cliente: Lista prod. nº:
Pedido nº: Cantidad:
Lote c/stock (nº/fecha):
Código:



ESPECIFICACIONES PARTICULARES

Acabado: x
Desarrollo: x
Mano izquierda, cantidad:
PL:
Mano derecha, cantidad:
Comentarios:

Modificaciones Modifications				Fecha Date	Nombre Name	Firma Signature	Material/Material:
Alt. Hgt.	Fecha Date	Nom. Name	Dibujado Drawn:	13/12/10	fgracia		Acabado/Finished:
			Comp. Comp:	13/12/10	xribera		
			Escala: Scale: %				

TECNOLAMA

Proy./Proj.: CSTD - CSCH
Plano/Drawing: E120-C2
Hoja/Page: 12/12
Programa/Program: ------
Código/Code:

E120:
TAPAJUNTAS
ADJUSTMENTS

RELLANO
LANDING DOOR

T2